Received: 26 April 2018 | Revised: 26 October 2018 | Accepted: 19 November 2018

DOI: 10.1002/nag.2918

RESEARCH ARTICLE

WILEY

A generic approach to modelling flexible confined boundary conditions in SPH and its application

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Funding information

Australian Research Council, Grant/Award Numbers: DP160100775, DP170103793 and FT140100408

Summary

In this paper, a new approach to applying confining stress to flexible boundaries in the smoothed particle hydrodynamics (SPH) method is developed to facilitate its applications in geomechanics. Unlike the conventional SPH methods that impose confining boundary conditions by creating extra boundary particles, the proposed approach makes use of kernel truncation properties of SPH approximations that occur naturally at free-surface boundaries. Therefore, it does not require extra boundary particles and, as a consequence, can be utilised to apply confining stresses onto any boundary with arbitrary geometry without the need for tracking the curvature change during the computation. This enables more complicated problems that involve moving confining boundaries, such as confining triaxial tests, to be simulated in SPH without difficulties. To further enhance SPH applications in elasto-plastic computations of geomaterials, a robust numerical procedure to implement Mohr-Coulomb plasticity model in SPH is presented for the first time to avoid difficulties associated with corner singularities in Mohr-Coulomb model. The proposed approach was first validated against two-dimensional finite element (FE) solutions for confining biaxial compression tests to demonstrate its predictive capability at small deformation range when FE solutions are still valid. It is then further extended to three-dimensional conditions and utilised to simulate triaxial compression experiments. Simulation results predicted by SPH show good agreement with experiments, FE solutions, and other numerical results available in the literature. This suggests that the proposed approach of imposing confining stress boundaries is promising and can handle complex problems that involve moving confining boundary conditions.

KEYWORDS

confining boundary, smoothed particle hydrodynamics (SPH), strain localisation

1 | INTRODUCTION

The confined boundary condition is commonly used in both experimental and numerical investigations of the mechanical behaviour of geomaterials, such as soils, rocks, and concretes; to replicate the in situ stress conditions, these materials are subjected to in the field.[1,2] Two categories of mechanical boundary conditions, which are velocity/displacement and stress/force, are commonly enforced to a specimen. In the laboratory, velocity and stress can be applied to the specimen through moving pistons or pressurised cells controlled by a feedback system consisting of a series of sensors and gauges.[3-5] In contrast, in numerical modelling, the ways to enforce such mechanical boundaries vary among different methods. For instance, in the finite element method (FEM), which has been regarded as the standard tool in continuum plasticity manifesting high stability and accuracy for applications involved small deformation, the boundary conditions can be directly applied to nodes on the surface meshes. In much well-developed commercial software such as ABAQUS, built in options allow these boundary conditions to be enforced in a straightforward manner. However, it is well established that the FEM may suffer from mesh pathologies such as mesh distortion and dependency, which hinders its applications to large deformation problems commonly encountered in engineering practice.[6,7] Although enhancements such as adaptive remeshing,[8] updated mesh,[9] hybrid Euler-Lagrangian method,[10] and smoothed FEM (SFEM)smoothed FEM (SFEM)[11] have been proposed, mesh-based numerical methods are still vulnerable to very large or discontinuous deformation involved in post-failure. On the other hand, mesh-free continuum methods are considered alternatives to FEM, for instance, the material point method (MPM),[12] and smoothed particle hydrodynamics (SPH).[13,14] In MPM, essential boundaries are enforced by replacing the interpolated velocities in the background mesh with predefined boundary values, while stress can be added to the surface of the computational domain in a similar way.[15] However, boundaries of the MPM particle assembly have to be aligned with the background mesh in order for the boundary conditions to be implemented. This procedure can become complicated and result in significant inaccuracy with highly deformed boundaries or in three-dimensional space.[16] It is clear that the above issues with FEM and MPM are collectively related to mesh discretisation. Consequently, as a truly meshless method, SPH is a good alternative to MPM for modelling large deformation problems. SPH was originally proposed for astrophysical applications[13,14]; it has been then advanced to numbers of applications in geomechanics such as large deformation and failure of geomaterials[17-24], slope failures and landslides,[25-27]) soil-structure interaction[28-31], coupled soil-water problems[18,32,33], and most recently the scale-dependent rock fracture.[34,35] In SPH, velocity and displacement boundary conditions can be enforced through non-slip and free-slip solid boundary conditions by using ghost[36] and virtual[19,37,38] particles. On the other hand, the stress-type boundary condition can be directly applied to SPH particles at the desired location where the confining stress is imposed. In such an approach, the applied stress should be converted to an equivalent acceleration, which can be subsequently added to the motion equation of each particle. The area over which the stress is applied and the corresponding particle masses are tracked in this approach. Furthermore, it requires the calculation of normal directions on boundary surfaces to apply the stresses correctly. Accordingly, a successful implementation of this approach requires (1) tracking of particles on the boundaries and (2) calculation of the normal vector and surface area to convert stresses to accelerations during the computation. However, these requirements would become extremely difficult to be satisfied when the computational domain undergoes large deformation and failure. This is because the highly curved boundary surfaces encountered in the large deformation of the numerical specimen could hinder the detection of the particles located on the surface boundaries and thus the calculation of their normal vectors. In addition, an accurate enforcement of the stress on the computational domain is challenging due to the difficulties in determining and tracking the surface area and curvature of the boundaries during the simulation. Although methods utilising kernel gradient to derive the normal direction on SPH domain surface is proposed to simplify the above process,[39,40] there is still a need to accurately calculate surface area of boundary domain over which the confining stresses are applied, and this still remains a challenging task for SPH when dealing with large deformation problems in solid mechanics.

To overcome the above issues, in this paper, a generic approach to applying confining stress to flexible boundaries is proposed for SPH. In contrast with previous approaches which are vulnerable to deformation of the confined boundaries, the current method automatically traces curvature change of the computational domain and enforces confining stress without being vulnerable to large deformation. In particular, the proposed method takes advantage of the kernel truncation near the boundaries of SPH domain. Such truncation in the kernel supporting domain would highlight the location of surface boundaries and allow accurate calculations of their normal directions as well as surface areas. As such, the proposed confining boundary condition outperforms the conventional methods in the following aspects: first, the boundaries of the computational domain are automatically located and their curvature change is accounted for without explicitly involving any searching process; second, the normal directions along the boundaries are automatically determined by the kernel truncation; and finally, the surface boundaries over which the confining stress is applied can be automatically determined during the computation. The above features can be automatically achieved through kernel approximation of the governing equation of the proposed method without explicitly calculating the area and normal on boundaries. These features enable the proposed method to perform effectively and accurately in enforcing confining pressures onto the computational domain. In addition, to further enhance the application of SPH to computational geomechanics, a modern elastoplastic model is required. For instance, the modified Cam-Clay,[41] Val-Eekelen,[42] Matsuoka-Nakai,[43] and Lade-Duncan[44] models enable the solution of complicated boundary value problems. Other more advanced models that can handle both diffuse and localised failure at the constitutive level have also been explored in the SPH framework for modelling fracture,[35] while further developments for applications in geomechanics are underway.[45-48] Despite this fact, the classical Mohr-Coulomb model is used here considering its simplicity in practical applications given the focus on theoretical work and corresponding algorithms for applying stress boundary conditions in SPH. For this, a robust numerical procedure to implement this model in SPH is presented for the first time to avoid difficulties associated with its well-known corner singularity. The Mohr-Coulomb–based constitutive models have been long applied and widely recognised in geomechanics to describe soil behaviour. Despite the popularity, this model possesses hexagon-shaped yield surface, which could lead to difficulties in modelling large deformation plasticity problems due to a discontinuous surface gradient. In this study, such instability is mitigated with a smooth approximation of the sharp corners using the Drucker-Prager yield surface, which features circular shape. The proposed approach is then validated against finite element solutions for biaxial and triaxial tests, GIMPM solutions by Kiriyama,[49] and triaxial compression experiment.[49]

The rest of this article is arranged as follows: The basic SPH concepts and governing equations are first presented. This is then followed by the general description of the numerical procedure to implement the Mohr-Coulomb elastic-perfectly plastic constitutive model in SPH. Next, the generic approach to applying confining boundary conditions in SPH is explained. Finally, the verification and application of the proposed framework are carried out to demonstrate the key features and potential of the proposed approach.

2 | SPH GOVERNING EQUATIONS FOR SOIL

2.1 | Basic SPH

In the SPH method, a continuum domain is discretised into an assembly of particles, each of which possesses field variables (such as mass, density, and stress) and moves with its own velocity. The field variables are then calculated through a kernel interpolation process, which can be expressed using the following equation[50]:

$$f(\mathbf{r}) = \int_{\Omega} f\left(\mathbf{r}'\right) W\left(\mathbf{r}-\mathbf{r}', h\right) \mathrm{d}\mathbf{r}', \tag{1}$$

where $f(\mathbf{r})$ represents the field variable at location $\mathbf{r}$, $f(\mathbf{r}')$ is the surrounding field variable at $\mathbf{r}'$ within the integral domain Ω, and W is the kernel function with the smoothing length (h) defining the size of Ω.

The selection of the kernel function can affect the accuracy and stability of numerical integration. In this paper, the cubic spline function[51] is adopted and has the following form:

$$W(q,h) = \alpha_d \times \begin{cases} 1-\dfrac{3}{2}q^2+\dfrac{3}{4}q^3 & 0 \le q < 1 \\ \dfrac{1}{4}(2-q)^3 & 1 \le q < 2 \\ 0 & q \ge 2 \end{cases}, \tag{2}$$

where α_d is a normalisation coefficient, which is equal to $10/(7\pi h^2)$ and $3/(2\pi h^3)$ in two- and three-dimensional space, respectively; q is the ratio between $|\mathbf{r}-\mathbf{r}'|$ and h.

In SPH, the kernel interpolation (1) is further discretised into a finite set of particles by replacing the integral with a summation. Accordingly, the field variables of a particle i can be approximated using the following equation:

$$f(\mathbf{r}_i) \approx \sum_{j=1}^{N} \frac{m_j}{\rho_j} f\left(\mathbf{r}_j\right) W\left(\mathbf{r}_i-\mathbf{r}_j, h\right), \tag{3}$$

where N is the total number of neighbouring particles located within the kernel support domain of particle i and m_j and ρ_j are the mass and density of the surrounding particles, respectively.

The SPH approximation for the gradient of a field function can be obtained by taking the derivative of Equation 1 and making use of the summation approximation in a similar way to the derivation of Equation 3. This leads to the following equation:

$$\frac{\partial f(\mathbf{r}_i)}{\partial \mathbf{r}_i} \approx \sum_{j=1}^{N} \frac{m_j}{\rho_j} f(\mathbf{r}_j) \frac{\partial W(\mathbf{r}_i - \mathbf{r}_j, h)}{\partial \mathbf{r}_i}. \tag{4}$$

Finally, by introducing the following notations:

$$W_{ij} = W(|\mathbf{r}_i - \mathbf{r}_j|, h) \qquad and \qquad \frac{\partial W_{ij}}{\partial \mathbf{r}_i} = \frac{\partial W_{ij}}{\partial \mathbf{r}_i} \frac{\partial \mathbf{r}_i}{|\mathbf{r}_i|}, \tag{5}$$

the expression of kernel interpolation in SPH can be simplified as follows:

$$f(\mathbf{r}_i) \approx \sum_{j=1}^{N} \frac{m_j}{\rho_j} f(\mathbf{r}_j) W_{ij}, \tag{6}$$

$$\frac{\partial f(\mathbf{r}_i)}{\partial \mathbf{r}_i} \approx \sum_{j=1}^{N} \frac{m_j}{\rho_j} f(\mathbf{r}_j) \frac{\partial W_{ij}}{\partial \mathbf{r}_i}. \tag{7}$$

Equations 3 and 4, and thus (6) and (7), can theoretically have second-order accuracy. However, this is not always achieved in SPH approximations, especially when particles undergo large deformation or the kernel supporting domain is truncated by boundaries. These deficiency problems are often called particle inconsistency and have been extensively studied in the past few decades. Different correction techniques have been proposed to restore particle consistency, thereby improving the accuracy of SPH approximations.[52-55] In this study, since the SPH approximation of the gradient of a function is mostly used to discretise the governing equations and to enforce confining boundary conditions, the accurate evaluation of the SPH approximation of the kernel gradient is essential. To achieve this, the SPH renormalisation technique[52,55,56] is adopted, which modifies the conventional SPH approximation of function gradient, ie, Equation 7, as follows:

$$\frac{\partial f(\mathbf{r}_i)}{\partial \mathbf{r}_i} \approx \sum_{j=1}^{N} \frac{m_j}{\rho_j} f(\mathbf{r}_j) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i}, \tag{8}$$

where $\partial W_{ij}^R / \partial \mathbf{r}_i = L_i \partial W_{ij} / \partial \mathbf{r}_i$ denotes the normalised kernel gradient at particle i and L_i is the renormalisation matrix, which has the following discretised form:

$$L_i = \begin{bmatrix} \sum_{j=1}^{N} \frac{m_j}{\rho_j}(x_j - x_i)\frac{\partial W_{ij}}{\partial x_i} & \sum_{j=1}^{N} \frac{m_j}{\rho_j}(x_j - x_i)\frac{\partial W_{ij}}{\partial y_i} & \sum_{j=1}^{N} \frac{m_j}{\rho_j}(x_j - x_i)\frac{\partial W_{ij}}{\partial z_i} \\ \sum_{j=1}^{N} \frac{m_j}{\rho_j}(y_j - y_i)\frac{\partial W_{ij}}{\partial x_i} & \sum_{j=1}^{N} \frac{m_j}{\rho_j}(y_j - y_i)\frac{\partial W_{ij}}{\partial y_i} & \sum_{j=1}^{N} \frac{m_j}{\rho_j}(y_j - y_i)\frac{\partial W_{ij}}{\partial z_i} \\ \sum_{j=1}^{N} \frac{m_j}{\rho_j}(z_j - z_i)\frac{\partial W_{ij}}{\partial x_i} & \sum_{j=1}^{N} \frac{m_j}{\rho_j}(z_j - z_i)\frac{\partial W_{ij}}{\partial y_i} & \sum_{j=1}^{N} \frac{m_j}{\rho_j}(z_j - z_i)\frac{\partial W_{ij}}{\partial z_i} \end{bmatrix}^{-1}. \tag{9}$$

2.2 | Governing equations

The deformation of an infinitesimal volume element within a continuum domain can be described using two fundamental governing laws, the mass and momentum conservation equations, which are written as follows:

$$\frac{D\rho}{Dt} = -\rho \frac{\partial v^{\alpha}}{\partial r^{\alpha}}, \tag{10}$$

$$\frac{Dv^{\alpha}}{Dt} = \frac{1}{\rho}\frac{\partial \sigma^{\alpha\beta}}{\partial r^{\beta}} + f^{\alpha}, \tag{11}$$

where α and β denote x, y and z axes of the Cartesian coordinate system with Einstein's convention applied to the repeated indices; ρ is the material density, v^{α} is the velocity component; $\sigma^{\alpha\beta}$ is the stress tensor component, taken negative in compression; f^{α} is the acceleration due to external loads such as gravitational force.

By using the SPH approximation of the gradient of a function, ie, Equation 8, the above partial differential governing equations can be discretised as follows:

$$\frac{D\rho_i}{Dt} = \sum_{j=1}^{N} m_j \left(v_i^{\alpha} - v_j^{\alpha} \right) \frac{\partial W_{ij}^{\mathrm{R}}}{\partial r_i^{\alpha}}, \tag{12}$$

$$\frac{Dv_i^{\alpha}}{Dt} = \sum_{j=1}^{N} m_j \left(\frac{\sigma_i^{\alpha\beta} + \sigma_j^{\alpha\beta}}{\rho_i \rho_j} + C_{ij}^{\alpha\beta} \right) \frac{\partial W_{ij}^{R}}{\partial r_i^{\beta}} + f_i^{\alpha}, \tag{13}$$

where $C_{ij}^{\alpha\beta}$ is a stabilising term consisting of artificial viscosity, which is utilised to remove numerical instability issues associated with SPH method,[25] and f_i^{α} is the force per unit mass due to gravitation. Finally, to solve the above system of governing equations, one needs a constitutive relation to calculate the stress tensor of the soil and this will be explained in the next section.

3 | CONSTITUTIVE MODEL

3.1 | Mohr-coulomb elasto-plastic model

The Mohr-Coulomb elasto-plastic model is widely used in practice owing to its simplicity in specifying the soil constitutive parameters.[57-59] In particular, this model requires five basic soil parameters including elastic modulus, Poisson's ratio, friction angle, dilation angle, and cohesion, which can be obtained from basic soil experiments such as triaxial or direct shear tests, thus making this model more appealing in practical applications. In this paper, a robust numerical framework to implement the Mohr-Coulomb constitutive model in SPH is presented in a pointwise manner following the SPH framework developed by Bui et al.[19] The framework makes use of the midpoint integration method to avoid iterative procedure, thus reducing computational time and providing the necessary accuracy of the solution within small strain increment.[60] The method is well suitable for incorporation in SPH that makes use of the Leap-frog time integration scheme (or explicit midpoint rule) to update the governing equations. We note that other schemes such as generalised trapezoidal rule,[61] higher order Runge-Kutta,[62] predictor-correct,[63] and backward Eulerian[64] are equivalently applicable.

To expand the detail of this model, we start with the definition of total strain rate tensor, which can be divided into elastic and plastic parts, as follows:

$$\dot{\varepsilon}^{\alpha\beta} = \dot{\varepsilon}_e^{\alpha\beta} + \dot{\varepsilon}_p^{\alpha\beta}, \tag{14}$$

where the subscripts e and p stand for elastic and plastic components, respectively.

The elastic strain rate tensor can be described by generalised Hooke's law:

$$\dot{\varepsilon}_{\mathrm{e}}^{\alpha\beta} = \frac{\dot{s}^{\alpha\beta}}{2G} + \frac{1}{9K}\dot{\sigma}^{\gamma\gamma}\delta^{\alpha\beta}, \tag{15}$$

where $\dot{s}^{\alpha\beta}$ is the deviatoric stress rate tensor; $G$and K are the material shear and bulk modulus; $\delta^{\alpha\beta}$ is the Kronecker's delta.

The plastic strain rate tensor is calculated using the plastic flow rule:

$$\dot{\varepsilon}_{\mathrm{p}}^{\alpha\beta} = \dot{\lambda}\frac{\partial g}{\partial \sigma^{\alpha\beta}}, \tag{16}$$

where $\dot{\lambda}$ is the time rate of the plastic multiplier; g is the plastic potential function, which defines the direction of plastic strain increment on the yield surface.

By substituting Equations 15 and 16 into Equation 14 and rearranging the obtained expression, the generalised stress–strain relationship of an elastic-perfectly plastic material can be written as follows:

$$\dot{\sigma}^{\alpha\beta} = 2G\dot{e}^{\alpha\beta} + K\dot{\varepsilon}^{\gamma\gamma}\delta^{\alpha\beta} - \dot{\lambda}\left[\left(K - \frac{2G}{3}\right)\frac{\partial g}{\partial\sigma^{mn}}\delta^{mn}\delta^{\alpha\beta} + 2G\frac{\partial g}{\partial\sigma^{\alpha\beta}}\right], \tag{17}$$

where $\dot{e}^{\alpha\beta} = \dot{\varepsilon}^{\alpha\beta} - \frac{1}{3}\dot{\varepsilon}^{\gamma\gamma}\delta^{\alpha\beta}$ is the deviatoric strain rate tensor; K is the elastic bulk modulus; m and n are free indexes, which are independent from α and β.

Next, the general formula of the plastic multiplier for the elastic-perfectly plastic model can be derived from the consistency condition, which requires the following:

$$\mathrm{d}f = \frac{\partial f}{\partial\sigma^{\alpha\beta}}\mathrm{d}\sigma^{\alpha\beta} = 0. \tag{18}$$

By substituting Equations 17 into 18, the general form of the time rate of the plastic multiplier for an elastic-perfectly plastic material can be written as follows:

$$\dot{\lambda} = \frac{2G\dot{\varepsilon}^{\alpha\beta}\dfrac{\partial f}{\partial\sigma^{\alpha\beta}} + \left(K - \dfrac{2G}{3}\right)\dot{\varepsilon}^{\gamma\gamma}\dfrac{\partial f}{\partial\sigma^{\alpha\beta}}\delta^{\alpha\beta}}{2G\dfrac{\partial f}{\partial\sigma^{mn}}\dfrac{\partial g}{\partial\sigma^{mn}} + \left(K - \dfrac{2G}{3}\right)\dfrac{\partial f}{\partial\sigma^{mn}}\delta^{mn}\dfrac{\partial g}{\partial\sigma^{mn}}\delta^{mn}}. \tag{19}$$

The Mohr-Coulomb yield surface and its plastic potential function can now be introduced and are expressed in the pressure-dependent form as follows, respectively:

$$f = \sin\phi I_1 + \frac{1}{2}\left[3(1 - \sin\phi)\sin\theta + \sqrt{3}(3 + \sin\phi)\cos\theta\right]\sqrt{J_2} - 3c\cos\phi = 0, \tag{20}$$

$$g = \sin\psi I_1 + \frac{1}{2}\left[3(1 - \sin\psi)\sin\theta + \sqrt{3}(3 + \sin\psi)\cos\theta\right]\sqrt{J_2} - 3c\cos\psi, \tag{21}$$

where ϕ and ψ are the soil internal friction and dilatant angles, respectively; $I_1 = \sigma^{\alpha\beta}\delta^{\alpha\beta}$, $J_2 = \frac{1}{2}s^{\alpha\beta}s^{\beta\alpha}$ and $J_3 = \frac{1}{3}s^{\alpha\beta}s^{\beta m}s^{m\alpha}$ are the first principal, second, and third deviatoric stress invariants, respectively; c is soil cohesion; and θ is the Lode angle defined as follows:

$$\theta = \frac{1}{3}\cos^{-1}\left(1.5\sqrt{3}\frac{J_3}{J_2^{1.5}}\right). \tag{22}$$

By substituting 20 and 21 into 19, the general form of the plastic multiplier reads the following:

$$\dot{\lambda} = \frac{1}{H}\left[3K\frac{\partial f}{\partial I_1}\dot{\varepsilon}^{\gamma\gamma} + 2G\left(\frac{\partial f}{\partial J_2}s^{\alpha\beta} + \frac{\partial f}{\partial J_3}t^{\alpha\beta}\right)\dot{\varepsilon}^{\alpha\beta}\right], \tag{23}$$

where H and $t^{\alpha\beta}$ are defined as follows:

$$H = 9K\frac{\partial f}{\partial I_1}\frac{\partial g}{\partial I_1} + 4GJ_2\frac{\partial f}{\partial J_2}\frac{\partial g}{\partial J_2} + 6GJ_3\frac{\partial f}{\partial J_2}\frac{\partial g}{\partial J_3} + 6GJ_3\frac{\partial g}{\partial J_2}\frac{\partial f}{\partial J_3} + 2G\left(s^{\alpha m}s^{m\beta}s^{\alpha n}s^{n\beta} - \frac{4}{3}J_2{}^2\right)\frac{\partial f}{\partial J_3}\frac{\partial g}{\partial J_3}, \tag{24}$$

$$t^{\alpha\beta} = \frac{\partial J_3}{\partial\sigma^{\alpha\beta}} = s^{\alpha m}s^{m\beta} - \frac{2}{3}J_2\delta^{\alpha\beta}. \tag{25}$$

In the above equations, the differentiation of the yield function f against the stress invariants I_1, J_2 and J_3 are listed below:

$$\frac{\partial f}{\partial I_1} = \sin\phi, \tag{26}$$

$$\frac{\partial f}{\partial J_2} = \frac{1}{4\sqrt{J_2}}\left[3(1-\sin\phi)\sin\theta + \sqrt{3}(3+\sin\phi)\cos\theta\right] + \frac{3\sqrt{3}J_3}{8J_2^2\sin 3\theta}\left[3(1-\sin\phi)\cos\theta - \sqrt{3}(3+\sin\phi)\sin\theta\right], \tag{27}$$

$$\frac{\partial f}{\partial J_3} = -\frac{\sqrt{3}}{4J_2\sin 3\theta}\left[3(1-\sin\phi)\cos\theta - \sqrt{3}(3+\sin\phi)\sin\theta\right]. \tag{28}$$

The differentiation of the plastic potential function g with respect to the stress invariants are similarly defined by replacing the friction angle ϕ with the dilation angle ψ in Equations 26, 27, and 28. Furthermore, to maintain the objectivity of the constitutive model under large deformation, the Jaumann stress rate tensor is adopted. The final stress-strain relation for Mohr-Coulomb elastic-perfectly plastic constitutive model can be now expressed as follows:

$$\dot{\sigma}^{\alpha\beta} = \sigma^{\alpha\gamma}\dot{\omega}^{\beta\gamma} + \sigma^{\gamma\beta}\dot{\omega}^{\alpha\gamma} + 2G\dot{e}^{\alpha\beta} + K\dot{\varepsilon}^{\gamma\gamma}\delta^{\alpha\beta} - \dot{\lambda}\left[3K\sin\varphi\delta^{\alpha\beta} + 2G\left(\frac{\partial g}{\partial J_2}s^{\alpha\beta} + \frac{\partial g}{\partial J_3}t^{\alpha\beta}\right)\right], \tag{29}$$

where $\dot{\varepsilon}^{\alpha\beta}$ and $\dot{\omega}^{\alpha\beta}$ are the strain and spin rate tensors, which can be related to the gradient of the velocity as follows:

$$\dot{\varepsilon}^{\alpha\beta} = \frac{1}{2}\left(\frac{\partial v^{\alpha}}{\partial r^{\beta}} + \frac{\partial v^{\beta}}{\partial r^{\alpha}}\right), \tag{30}$$

$$\dot{\omega}^{\alpha\beta} = \frac{1}{2}\left(\frac{\partial v^{\alpha}}{\partial r^{\beta}} - \frac{\partial v^{\beta}}{\partial r^{\alpha}}\right). \tag{31}$$

It is noted that the above application of Jaumann rate could lead to nonphysical oscillations in some applications. However, in the current work, since the entire SPH governing equations are solved in an updated Lagrangian frame with a very small time step, the issues associated with the Jaumman rate can be significantly reduced (if not noticeable). This has been proven in simulations of granular flow with a good stability.[19,48] Nevertheless, we acknowledge some existing state-of-art stress rates, such as Green-Naghdi,[65] Truesdell,[66] Oldroyd,[67] and logarithmic,[68] can be also adopted to maintain objective stress rates.

3.2 | Singularity treatment and stress return algorithms

Figure 1 shows the Mohr-Coulomb yield surface in the π-plane and a treatment algorithm for its corner singularity problem. It can be seen from this figure that the yield and plastic potential functions are nondifferentiable on the vertices ($\theta = 0^{o}$ or 60^{o}), which corresponds to the stress state lying either on the tensile or compressive meridian of the

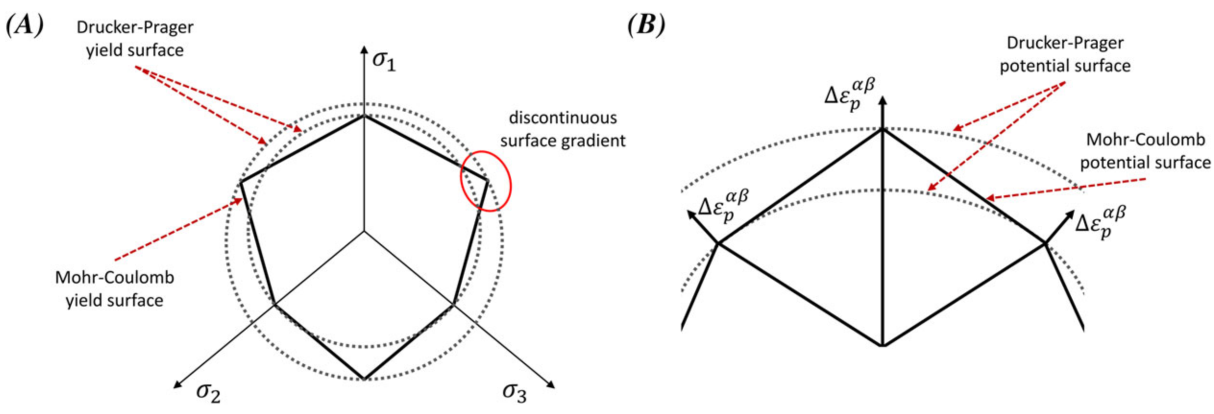

FIGURE 1 Illustration of the singularity problem in Mohr-Coulomb model: A, Mohr-Coulomb and Drucker Prager yield surfaces in π-plane. B, Treatment for the singularity problem in Mohr-Coulomb model [Colour figure can be viewed at wileyonlinelibrary.com]

Mohr-Coulomb potential surface. Many numerical algorithms have been proposed and demonstrated to be effective for treating this singularity problem. Examples of these algorithms are the smoothing of Mohr-Coulomb yield surface corners,[60] linearisation of the yield surface,[69] hyperbolic approximation of the sharp corners,[57] C1 and C2 continuous treatment of the model.[70] In this work, the approach proposed by Chen and Mizuno[60] is adapted for implementing the Mohr-Coulomb model in SPH. This approach is featured with a relatively straightforward implementation procedure without compromising accuracy as Mohr-Coulomb and Drucker-Prager surfaces coincide where surface gradient is discontinuous. The basic idea of this approach is to use the Drucker-Prager yield and potential functions to replace the Mohr-Coulomb ones when the stress state coincides with tensile or compressive meridian of the Mohr-Coulomb model (Figure 1B).

To facilitate the implementation, the Mohr-Coulomb yield and plastic potential functions can be rewritten in the following forms:

$$f = \alpha_\phi I_1 + \sqrt{J_2} - k_\phi, \tag{32}$$

$$g = \alpha_\psi I_1 + \sqrt{J_2} - k_\psi, \tag{33}$$

where α_ϕ and k_ϕ are calculated by fitting Equations 32 and 33 to Equations 20 and 21, which leads to the following expression of α_ϕ and k_ϕ:

$$\alpha_\phi = \frac{2\sin\phi}{3(1-\sin\phi)\sin\theta + \sqrt{3}(3+\sin\phi)\cos\theta}, \tag{34}$$

$$k_\phi = \frac{6c\cos\phi}{3(1-\sin\phi)\sin\theta + \sqrt{3}(3+\sin\phi)\cos\theta}. \tag{35}$$

The coefficients α_ψ and k_ψ in Equation 33 are defined in a similar manner by replacing the friction angle ϕ with the dilation angle ψ in Equations 34 and 35.

By applying the above definitions, the stress-strain relationship and plastic multiplier for corner singularity treatment of the Mohr-Coulomb constitutive model can be calculated as follows:

$$\dot{\sigma}^{\alpha\beta} = \sigma^{\alpha\gamma}\dot{\omega}^{\beta\gamma} + \sigma^{\gamma\beta}\dot{\omega}^{\alpha\gamma} + 2G\dot{e}^{\alpha\beta} + K\dot{\varepsilon}^{\gamma\gamma}\delta^{\alpha\beta} - \dot{\lambda}\left[3K\alpha_\psi\delta^{\alpha\beta} + \left(G/\sqrt{J_2}\right)s^{\alpha\beta}\right], \tag{36}$$

$$\dot{\lambda} = \frac{3\alpha_\phi K\dot{\varepsilon}^{\gamma\gamma} + \left(G/\sqrt{J_2}\right)s^{\alpha\beta}\dot{\varepsilon}^{\alpha\beta}}{9\alpha_\phi K\alpha_\psi + G}. \tag{37}$$

The condition for using the above stress-strain relation depends on the value of the Lode angle. In the principal stress space of the Mohr-Coulomb yield surface, the Lode angle falls within the range $0 \leq \theta \leq \pi/3$. Accordingly, if the Lode angle is equal to either the higher or lower bound of this range, the corresponding stress tensor is located on either the tensile or compressive meridian, meaning that the singularity problem occurs. This condition is expressed by the following equation:

$$\left|\frac{3\sqrt{3}}{2}\frac{J_3}{J_2^{3/2}}\right| < 1. \tag{38}$$

Apart from the corner singularity, another issue in the numerical implementation of the Mohr-Coulomb model is related to the tension crack problem. It manifests itself as the stress state in the soil falling beyond the apex of the yield surface on $\left(-I_1, \sqrt{J_2}\right)$ plane, which is defined by the following condition:

$$-\alpha_\phi I_1 + k_\phi < 0. \tag{39}$$

When the above condition is met, the normal stresses can be adjusted to shift the hydrostatic pressure back to the apex of the yield surface in order to correct the tension crack issue:

$$\widetilde{\sigma}^{\alpha\beta} = \sigma^{\alpha\beta} - \frac{1}{3}\left(I_1 - \frac{k_\phi}{\alpha_\phi}\right)\delta^{\alpha\beta}, \tag{40}$$

where $\sigma^{\alpha\beta}$ and $\widetilde{\sigma}^{\alpha\beta}$ are the stress tensors before and after the correction, respectively.

In addition, due to the accumulation of computational errors during the simulation, the stress state of the model could be lying above the yield surface. This problem can be determined by the following condition:

$$-\alpha_\phi I_1 + k_\phi < \sqrt{J_2}. \tag{41}$$

Equation 41 violates the basic assumption in the plasticity theory, which requires the stress state to remain on the yield surface during plastic deformation. Thus, whenever Equation 41 is satisfied, the stress state is scaled back to the yield surface using the following equations:

$$\widetilde{\sigma}^{\alpha\beta} = r s^{\alpha\beta} + \frac{1}{3} I_1 \delta^{\alpha\beta}, \tag{42}$$

$$r = \frac{-\alpha_\phi I_1 + k_\phi}{\sqrt{J_2}}, \tag{43}$$

where r is a scaling factor. This procedure guarantees that the stress state during numerical integration is always on the Mohr-Coulomb yield surface.

4 | BOUNDARY CONDITIONS IN SPH

Like any other numerical methods, the treatment of boundary conditions in SPH is required to facilitate its applications to a wide range of engineering problems. In particular, when an SPH particle is close to the boundary, its kernel function is truncated, resulting in inaccurate approximations of field variables. To resolve this problem, ghost and virtual boundary particles[19,36,39,71] have been introduced to replace solid boundaries. Specific conditions are then enforced through these particles to achieve desirable boundary conditions (eg, non-slip, free-slip, or axis-symmetric conditions). However, the above approaches are not directly applicable to moving or flexible boundaries subjected to confining stress and therefore specific treatments are required in such cases.

4.1 | Conventional approach to apply confining stress in SPH

In engineering applications, confining stress is normally applied to open boundaries such as free-surfaces or confining membranes. To replicate this condition in SPH, the common procedure consists of three main steps (illustrated in Figure 2):

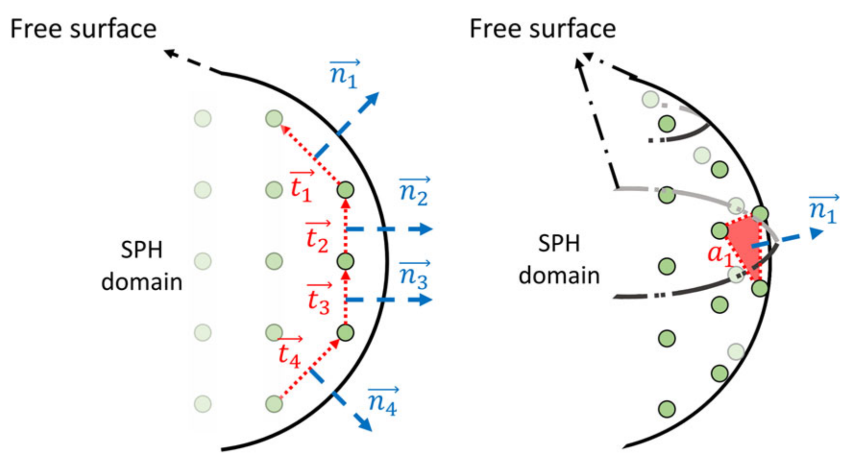

FIGURE 2 Conventional approach to apply confining stress in SPH (two-dimensional left and three-dimensional right) [Colour figure can be viewed at wileyonlinelibrary.com]

1. Identify particles on the open boundary where the confining stress is to be applied;
2. Calculate the normal vector for each particle on the open boundary; and
3. Calculate the surface area at each particle over which the confining stress is applied.

The above procedure was, in fact, proposed by Monaghan et al.[40] in extending SPH to model the motion of a rigid solid boundary in water. It was then further developed by Kajtar and Monaghan[72] to model swimming bodies. Recently, this approach has been applied to model confining stress in rectangular triaxial tests.[40]

To ensure the confining stress is correctly applied to the boundary, the above procedure has to be repeated at every integration step during the computation process, which is computationally expensive. Furthermore, when the computational domain undergoes large deformation, the normal direction and surface area cannot be accurately calculated, thus failing to maintain correct confining stress on the boundary and potentially leading to the termination of the computational procedure. As a consequence, basic soil experiments including biaxial and triaxial tests involving large deformation cannot always be modelled using this approach in SPH. Therefore, developing a new computational procedure is required to properly account for the confining stress in SPH modelling of soil plasticity problems.

4.2 | A new approach to applying confining stress to flexible boundaries in SPH—Theory

In this section, a generic approach to applying confining stress to flexible boundaries in SPH is developed. The proposed approach makes use of kernel truncation properties of SPH approximations near boundaries to facilitate the enforcement of confining stress on the boundaries, thereby allowing the extension of current SPH applications to more complex engineering problems that require confining stresses (such as biaxial or triaxial soil tests).

Let us first consider a dry solid body Ω of arbitrary shape subjected to a constant confining pressure field σ_c and is under equilibrium as shown in Figure 3. The stress state of any infinitesimal volume dV within Ω can be defined as follows:

$$\sigma^{\alpha\beta} = \sigma'^{\alpha\beta} + \sigma_c \delta^{\alpha\beta}, \tag{44}$$

where $\sigma'^{\alpha\beta}$ is the effective stress component due to the deformation of the solid skeleton and σ_c is the confining pressure acting on dV.

The motion of the infinitesimal volume dV within Ω can be then described by substituting Equation 44 into the SPH momentum Equation 13, as follows:

$$\frac{Dv_i^\alpha}{Dt} = \sum_{j=1}^{N} m_j \left(\frac{\sigma'^{\alpha\beta}_i + \sigma'^{\alpha\beta}_j}{\rho_i \rho_j} + \frac{\sigma_{ci} + \sigma_{cj}}{\rho_i \rho_j} \delta^{\alpha\beta} \right) \frac{\partial W_{ij}^R}{\partial r_i^\beta} + f_i^\alpha. \tag{45}$$

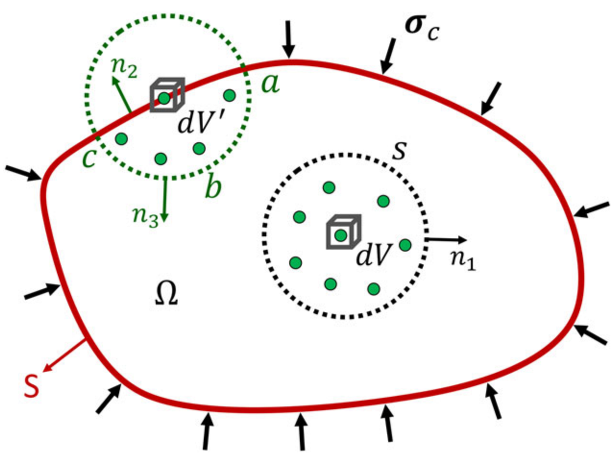

FIGURE 3 Illustration of the mechanism for the proposed confining boundary condition [Colour figure can be viewed at wileyonlinelibrary.com]

The above equation implies that any infinitesimal volume element dV located within Ω is subjected to a constant hydrostatic pressure component, which is equal to the constant applied confining stress σ_c. This suggests that the additional term associated with the confining stress in Equation 45 plays a role as the confining stress generator to the boundary surface of Ω, thereby producing a constant hydrostatic pressure everywhere within Ω. To demonstrate this, let us rewrite the term associated with the confining stress in Equation 45 in an integral form:

$$\sum_{j=1}^{N} m_j \left(\frac{\sigma_{ci} + \sigma_{cj}}{\rho_i \rho_j} \right) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} = \frac{1}{\rho_i} \int_{\Pi} (\sigma_{ci} + \sigma_{cj}) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} dV_j, \tag{46}$$

where dV_j is the volume of particle j or a volume element j, which is located within the supporting domain Π of i defined by the kernel function W_{ij}. The right hand-side of Equation 46 can be further extended as follows:

$$\frac{1}{\rho_i} \int_{\Pi} (\sigma_{ci} + \sigma_{cj}) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} dV_j = \frac{1}{\rho_i} \int_{\Pi} (\sigma_{cj} - \sigma_{ci}) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} dV_j + \frac{1}{\rho_i} \int_{\Pi} (2\sigma_{ci}) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} dV_j. \tag{47}$$

Under the equilibrium condition, the confining pressure is constant everywhere within the domain Ω, causing the first term of Equation 47 to vanish everywhere in Ω. Next, it will be proven that the second term of Equation 47 produces the confining stress on the boundary surface S. Using the divergence theorem, the second term on the right-hand side of Equation 47, which is a volume integral, can be converted into a surface integral as follows:

$$\frac{1}{\rho_i} \int_{\Pi} (2\sigma_{ci}) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} dV_j = -\frac{2\sigma_{ci}}{\rho_i} \left(\int_s W_{ij}^R \overrightarrow{n_1} ds \right), \tag{48}$$

where s is the surface of the kernel supporting domain Π and $\overrightarrow{n_1}$ is unit vector normal to s.

In SPH, the kernel function W is always defined to be symmetric and has a compacted supporting domain, thus the integral of $W\overrightarrow{n_1}$ (and thus $W_{ij}^R \overrightarrow{n_1}$) over any closed surface s is zero. This suggests that Equation 48 vanishes everywhere within Ω if s is a closed surface, or in other words, the supporting domain Π is not truncated. However, this is not the case when the volume element is located close to or on the surface boundary Ω. For this particular case, let us consider the volume element dV' as shown in Figure 3. The application of Equation 48 to this element leads to the following equation:

$$\frac{1}{\rho_i} \int_{\Pi} (2\sigma_{ci}) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} dV_j = -\frac{2\sigma_{ci}}{\rho_i} \left(\int_{ac} W_{ij}^R \overrightarrow{n_2} ds + \int_{abc} W_{ij}^R \overrightarrow{n_3} ds \right), \tag{49}$$

where $\overrightarrow{n_2}$ and $\overrightarrow{n_3}$ are the normal vectors on surface sections ac and abc (see Figure 3), respectively. Because abc is a closed surface, the surface integral of the kernel function $W_{ij}^R \overrightarrow{n_3}$ over this surface is zero, thus the above equation can be further simplified as below:

$$\frac{1}{\rho_i} \int_{\Omega} (2\sigma_{ci}) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} dV_j = -\frac{2\sigma_{ci}}{\rho_i} \int_{ac} W_{ij}^R \overrightarrow{n_2} ds. \tag{50}$$

The above equation integrates stress σ_c over the surface ac, which results in a force acting on the volume element dV' in the opposite direction to the normal vector $\overrightarrow{n_2}$. Consequently, when this surface integration is performed over the entire surface S of the domain Ω, it forms a constant confinement that is proportional to σ_c onto the surface area of the problem domain. Therefore, when applying Equation 45 to enforce confining stress, σ_c only takes effect on the free surface while automatically vanishing inside the domain. This suggests that the confining stress in SPH can be enforced through Equation 45 by assigning a constant pressure value, equal to the confining stress, to every SPH particle within the computational domain. This method provides a new pathway to effectively enforce a confining stress boundary condition in SPH without searching for particles located on or close to free-surfaces or open boundaries. Accordingly, for any problem under a constant confining stress, the following SPH motion equation can be applied with a constant confining pressure assigned to every SPH particle:

$$\frac{Dv_i^\alpha}{Dt} = \sum_{j=1}^{N} m_j \left(\frac{\sigma_i^{\alpha\beta} + \sigma_j^{\alpha\beta}}{\rho_i \rho_j} + C_{ij}^{\alpha\beta} \right) \frac{\partial W_{ij}^R}{\partial r_i^\beta} + \sum_{j=1}^{N} \frac{m_j}{\rho_i \rho_j} (\sigma_{ci} + \sigma_{cj}) \frac{\partial W_{ij}^R}{\partial r_i^\alpha} + f_i^\alpha. \tag{51}$$

4.3 | Verification of the proposed confining stress approach in SPH

In this section, an SPH benchmark test with a circularly shaped specimen subjected to a constant confining stress is carried out to verify the stability and accuracy of the proposed method to apply stress boundary condition in SPH. Figure 4 shows the outline of the model set-up for numerical tests. The test is conducted with a 10 kPa confining stress and a linear elastic constitutive model is assumed in the numerical simulation for simplicity. The material properties are $E = 20$ MPa and $\nu = 0.3$, while the initial interparticle distance was set to 1 mm. The sample has a radius of 50 mm, which is discretised using a fan shaped particle configuration resulting in a total of 1951 SPH particles as shown in Figure 4. Such fan shaped layout creates a smooth boundary surface compared to orthogonal particle layout, which can smear out numerical noise during simulation due to the zig-zag shape of the boundary. The confining stress is achieved by applying an initial hydrostatic pressure of $\sigma_c = 10$ kPa to every SPH particle together with the use of Equation 51 to describe the motion of SPH particles. Two numerical tests with and without an initial stress condition are conducted. In the former case, an initial stress condition that is equivalent to the applied confining stress of 10 kPa is assigned to every particle within the computational domain, and this can be achieved by setting $\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = 10$ kPa and $\sigma_{xy} = 0$ kPa to every SPH particle. On the other hand, all initial stress components are set to zero in the latter case to simulate the sample without an initial stress condition. Confining stress is enforced in both cases.

In addition, viscous damping is required in SPH to stabilise the potentially excessive stress wave propagation due to the sudden application of the confining stress vectors on the computational domain without an initial stress condition. This viscous damping can be incorporated in the SPH motion equation following the approach proposed by[26] as follows:

$$f_i^\alpha = f_i^\alpha - \xi \sqrt{\frac{E}{\rho h^2}} v_i^\alpha, \tag{52}$$

where f_i^α is the acceleration of particle i due to internal stress and external force; ξ is a dimensionless damping coefficient chosen to be 0.1,[26] (Nguyen et al. 2017).

Figure 5 shows the SPH simulation results for the case of 10 kPa confining pressure applied to the circular specimen using Equation 51. It can be seen that the proposed method can correctly produce the confining stress direction and its magnitude acting on the numerical specimen without any extra effort required to search for particles on the boundary and imposing the confining stress to these particles. The plots of confining stress vectors indicate that all stress components produced by the additional confining stress term in Equation 51 vanish within the numerical specimen except those on the boundary where the SPH kernel is truncated. The truncation of the SPH kernel approximation on the boundary produces stress vectors normal to it and directing towards the centre of the circular specimen, suggesting that the proposed method presented in the previous section is correct. By comparing the results between two cases, with and without imposing an initial confining stress of 10 kPa to the specimen, it shows that the latter case reaches an

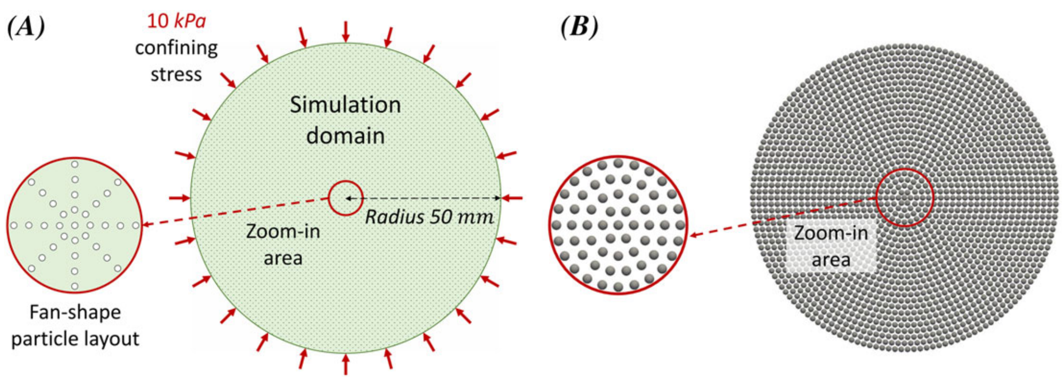

FIGURE 4 Model set-up for numerical test. A, Geometry of the problem. B, Radial particle layout in SPH simulation [Colour figure can be viewed at wileyonlinelibrary.com]

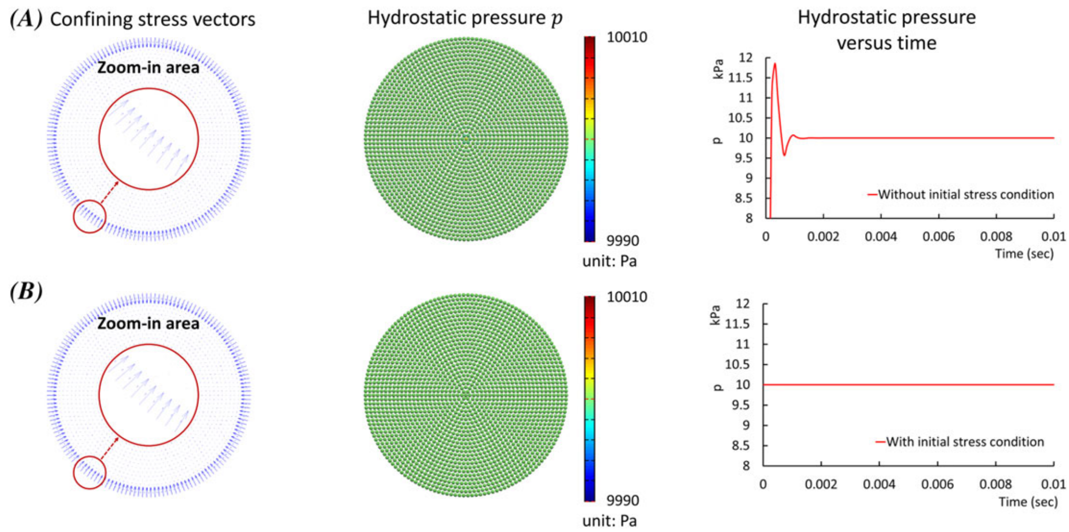

FIGURE 5 SPH simulation of confined compression test. A, Without imposing initial stress condition. B, With imposing initial stress condition [Colour figure can be viewed at wileyonlinelibrary.com]

equilibrium after several hundreds of cycles of numerical iterations with the aid of the viscous damping force, while the former case achieves an equilibrium immediately. Both cases result in the same initial confining stress of 10 kPa across the numerical specimen, suggesting that the additional confining stress term in Equation 51 can produce the confining stress that is of the same order of magnitude as with the initially imposed initial confining stress of 10 kPa to the numerical specimen. These good agreements also imply that the viscous damping has no influence on the simulation results but improves the stability of the numerical solutions.

The above numerical test indicates that the proposed method or Equation 51 can exactly produce the desirable confining stress to a computation domain by simply applying an initially uniform hydrostatic pressure to all SPH particles within the computational domain. No additional effort is required to search for particles on the confinement boundary and then impose the desirable confining stress to these particles. The proposed method can instantaneously achieve a desirable confining stress if a hydrostatic stress condition that is equivalent to the applied confining stress is imposed to all particles in the numerical specimen. Nevertheless, strictly speaking, the above numerical observation is only applicable for small deformation cases. In situations where large deformations occur, the relative movement of particles may affect the accuracy of the method. Therefore, further improvements are required to ensure stability and accuracy of the proposed method and these will be explained in the next section.

4.4 | Treatment of confining boundary condition for large deformations

When the material undergoes large deformations, SPH particles representing the computation domain no longer maintain their initial relative positions and can become highly disordered, which may hinder the application of the proposed confining stress approach. In particular, Equation 48 may become nonzero for those SPH particles undergoing very large deformation even though their supporting domain is still located within Ω (ie, theoretically remains a closed surface), causing an imbalance of confining forces acting on those SPH particles within the computational domain. One straightforward way to mitigate this issue in SPH applications involving very large soil deformation is to apply Equation 51 to only SPH particles which are located on or close to the boundary of Ω. The empirical condition to specify these SPH particles, which is optimised by trial and error, can be written as follows:

$$f_i = \begin{cases} \leq 0.55 & \text{in} \quad 2D \\ \leq 0.70 & \text{in} \quad 3D \end{cases}, \tag{53}$$

where f_i is an index value for a given SPH particle within the computational domain and it is calculated as follows:

$$f_i = \sum_{j=1}^{N} \frac{m_j}{\rho_j} W_{ij}, \tag{54}$$

with N being the total number of SPH particles located within the supporting domain of particle i and j representing these neighbouring particles.

On the other hand, for those SPH particles located within Ω where their supporting domain remains a closed surface, the second term in the right-hand side of Equation 51 vanishes and Equation 51 reduces to the standard SPH formulation for elasto-plastic materials:

$$\frac{Dv_i^{\alpha}}{Dt} = \sum_{j=1}^{N} m_j \left(\frac{\sigma_i^{\alpha\beta} + \sigma_j^{\alpha\beta}}{\rho_\mathrm{i}\rho_j} + C_{ij}^{\alpha\beta} \right) \frac{\partial W_{ij}^R}{\partial r_\mathrm{i}^{\beta}} + g_i^{\alpha}. \tag{55}$$

The above procedure guarantees that the confining stress vectors only take effect on the confinement boundary during the simulation and this is consistent with FEM and experimental results. It is worth emphasising here that, although an extra computational effort is required to identify SPH particles located on or close to the confinement boundaries (or free-surface boundaries), the current approach does not require the calculation of normal vectors and surface areas for particles over which the confining stress is applied. This will significantly reduce the overall computational effort required to apply the confining stress in SPH compared to the conventional approach as discussed in the preceding section.

In addition, owing to the fact that the smoothing length of SPH particles is kept constant in the current work, the interpolated value of the confining stress from Equation 48, or the second term on the right-hand side of Equation 51, may reduce when the particle separation increases. This may result in the reduction of the magnitude of the confining stress vectors and thus a loss of the required level of confinement on the modelling domain. One way to mitigate this issue is to update the smoothing length of SPH particles during the simulation.[73,74] Alternatively, the following simple approach can be applied to maintain the confining stress during the simulation. In particular, the second term on the right-hand side of Equation 51 is updated during the simulation using the following equation:

$$\sum_{j=1}^{N} \frac{m_j}{\rho_i\rho_j} (\sigma_{ci} + \sigma_{cj}) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} = \sum_{j=1}^{N} \frac{m_j}{\rho_i\rho_j} (\sigma_{ci} + \sigma_{cj}) \frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} * \left(\frac{l^0}{l^n} \right), \tag{56}$$

where n represents the nth time step; l^0 and l^n are the second term on the right-hand side of Equation 51 at the initial and nth time step, respectively, and can be expressed as follows:

$$l^n = \left\{ \sum_{j=1}^{N} \frac{m_j}{\rho_i\rho_j} (\sigma_{ci} + \sigma_{cj}) \left(\frac{\partial W_{ij}^R}{\partial \mathbf{r}_i} \right) \right\}^n. \tag{57}$$

5 | NUMERICAL VERIFICATION AND APPLICATION

5.1 | Performance of Mohr-Coulomb elasto-plasticity in SPH

An SPH simulation of a simple shear test using Mohr-Coulomb elasto-plastic constitutive model is conducted in this section, and its results are compared with theoretical solutions under plane strain condition.[75] The geometry and boundary conditions of a square shaped soil specimen of 0.3-m length are shown in Figure 6. An initial particle distance of 0.01 m is used, resulting in 900 particles for the specimen. Two areas are distinguished within the specimen: a centrally located area of 100-mm length where SPH particles move freely and a boundary area enclosing the central area where a constant velocity field is predefined and maintained during the simulation following Equation 58. The application of a predefined constant velocity field to the boundary area[76] maintains the pure shear condition and is suitable for testing constitutive models in SPH:

$$v_{xi} = 0.01 y_i, \tag{58}$$

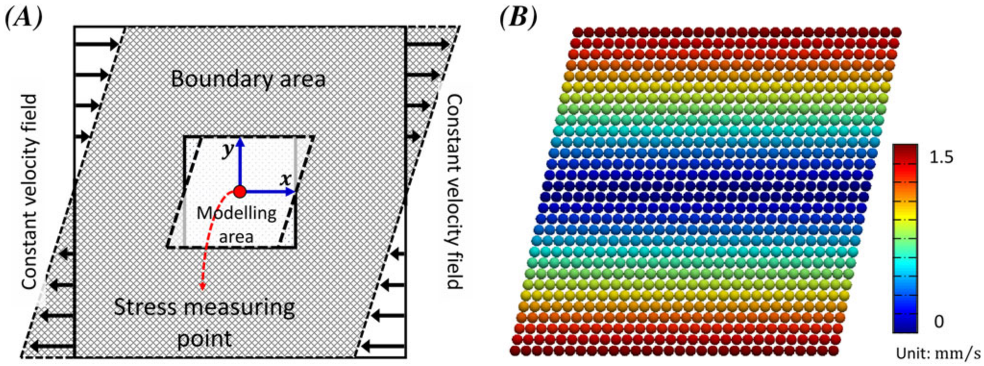

FIGURE 6 Simple shear test for Mohr-Coulomb model benchmark simulation: A, Simulation set-up. B, The contour plot of velocity [Colour figure can be viewed at wileyonlinelibrary.com]

where i indicates any particle within the boundary area. Three values of confining stress of 50, 75, and 100 kPa are tested to analyse the sample behaviour with its pressure dependent yield surface. The soil properties are: Young modulus $E = 20$ MPa, Poisson ratio $\nu = 0.3$, internal friction angle ?=30°, dilation angle $\psi = 0^\circ$, cohesion $c = 10$ kPa, and density $\rho = 2100$ kg/m^3.

The stress path and shear stress-strain relationships are obtained at the centre (Figure 7A) and plotted in Figure 7, while the magnitude of the velocity profile in SPH simulations is shown in Figure 6B. In Figure 7A, the stress states deviate from their initial hydrostatic confined states (150, 225, and 300 kPa) as the shear starts. The stress path is vertical, indicating that the confining stress is correctly maintained during the shear test, and the stress state remains on the yield surface, as a verification of the consistency condition being respected during loading. For shear stress-strain relations (Figure 7B), the numerical results agree well with theoretical calculations, showing pressure dependent solutions. The above results demonstrate a stable and accurate performance of the Mohr-Coulomb model incorporated in SPH, and its effective treatment to corner singularity problem.

5.2 | SPH simulation for biaxial test

Plane-strain biaxial tests are simulated in this section to further verify the proposed confining boundary condition and the Mohr-Coulomb elasto-plastic constitutive model in SPH. The obtained numerical results are then compared with those derived from the classical finite element method (ABAQUS) as well as the generalised interpolation material point method (GIMPM)[49] with elastoplastic Mohr-Coulomb model. The initial geometry setting and boundary conditions for the biaxial test are shown in Figure 8. The numerical sample has a rectangular shape with an initial dimension of 100-mm height and 50-mm width (Figure 8A). In the simulation, the numerical sample is first isotropically loaded to a predefined confining stress level (Figure 8B). The confining stress on the two side boundaries is then kept unchanged during the simulation, while the motions of top and bottom boundaries are controlled for axial loading. In this test, the motion along the bottom boundary is restrained in all directions, while that in the vertical, direction of the top boundary is subjected to a constant downward velocity of 10 mm/s. The horizontal motion along the top boundary is

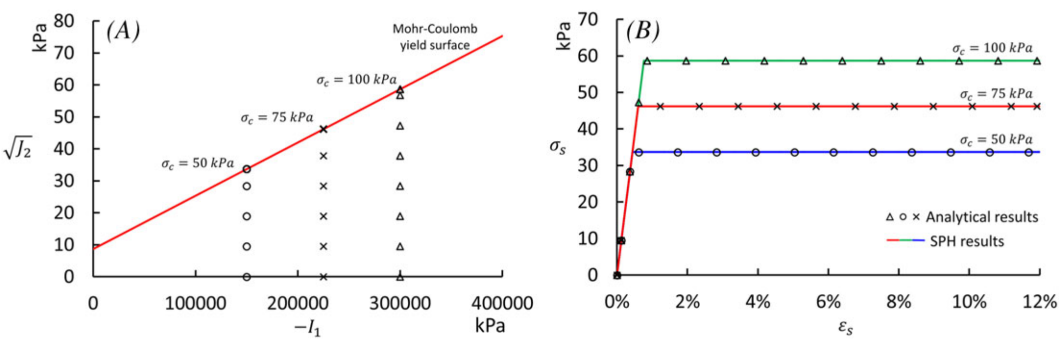

FIGURE 7 A, Loading path for varying confining stresses. B, Shear stress-strain relationships for varying confining stresses [Colour figure can be viewed at wileyonlinelibrary.com]

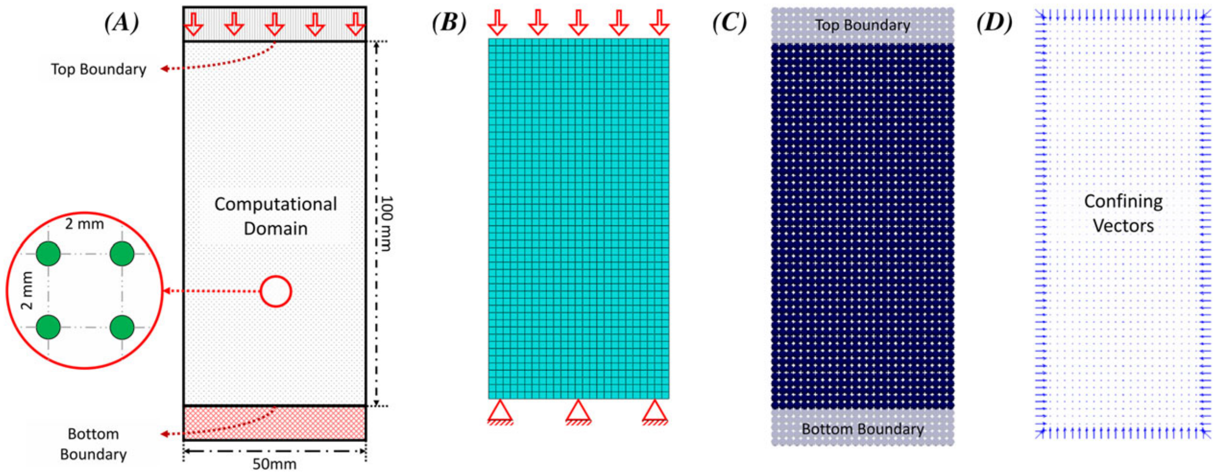

FIGURE 8 Initial setting geometry and boundary conditions used in simulations. A, Initial geometry for the biaxial test. B, FEM mesh discretisation. C, SPH particle discretisation. D, Confining stress vectors generated by Equation 56 [Colour figure can be viewed at wileyonlinelibrary.com]

controlled to replicate the free-cap boundary (ie, freely moved) and fixed-cap boundary (ie, restrained) conditions, which are commonly reported in experiments. The material properties are summarised in Table 1 along with their predefined confining stress levels.

In SPH simulations, the above numerical sample is created using 1250 particles, which are arranged in a square lattice with an initial lattice size of 2 mm and an initial smoothing length of 2.4 mm (Figure 8C). In addition, five extra layers of SPH particles are created at both the top and bottom boundary areas to enforce boundary conditions. The fixed boundary or constant velocity boundary in SPH can be readily achieved by restricting particle position update or restricting particle velocity update, while the confining stress is enforced through the proposed confining boundary approach described in the preceding section. All other parameters for SPH simulations are selected following the suggestion by Bui et al[19] except that the sound speed for the artificial viscosity is computed following the work of Bui et al.[25] In FEM, to provide a quantitative comparison with SPH, an initial mesh size of 2 mm is also adopted (Figure 8B). However, different from SPH that requires extra particles to impose the displacement controlled boundary condition, the boundary conditions in FEM can be straightforwardly achieved by directly applying predefined values to nodes located on boundary surfaces of the computational domain. These boundary conditions are applied to the numerical sample through the built-in options in ABAQUS. It is also noted here that a built-in option Nlgeom, which proceeds nonlinear deformation in computational domain, is activated to avoid ill-posed boundary values for all FEM simulations.

The development of deviatoric strain in SPH biaxial tests and their quantitative comparisons with FEM results are illustrated in Figure 9 with a horizontally restricted top boundary condition. The sample responses are tested with confining stresses of 50 and 100 kPa, respectively, and the deviatoric strain is calculated as $\varepsilon^{\text{dev}} = \left(\frac{1}{2}\boldsymbol{\epsilon}:\boldsymbol{\epsilon}\right)^{1/2}$, where $\boldsymbol{\epsilon}$ is the deviatoric strain tensor. From the results, SPH yields a very similar prediction of strain localisation pattern compared with the classical FEM solution, which demonstrates a good stability of current SPH framework when simulating soil plasticity problems. During the SPH test with 50-kPa confinement (Figure 9A), a uniform distribution of the accumulated deviatoric strain is observed in the sample at an early stage of loading. When the stress state reaches the yield surface and the strain field bifurcates, a symmetric cross-shaped localisation of strain field forms within the sample, which is observed at 0.9% axial deformation. As the deformation proceeds, deformation accumulates within the shear band and facilitates the localisation process. At 1.2% axial strain, the originally symmetric cross-shaped shear bands start to develop in a nonsymmetric manner, in which one branch of the cross shaped shear band develops more than the

TABLE 1 Model parameters for the biaxial tests

Test No.	Confining Stress, kPa	Shear Modulus, kPa	Density	Cohesion, g/cm^3	Friction Angle, °	Dilatant Angle, °
1	50	6401	1.53	8.5	30.5	0
2	100	8514				

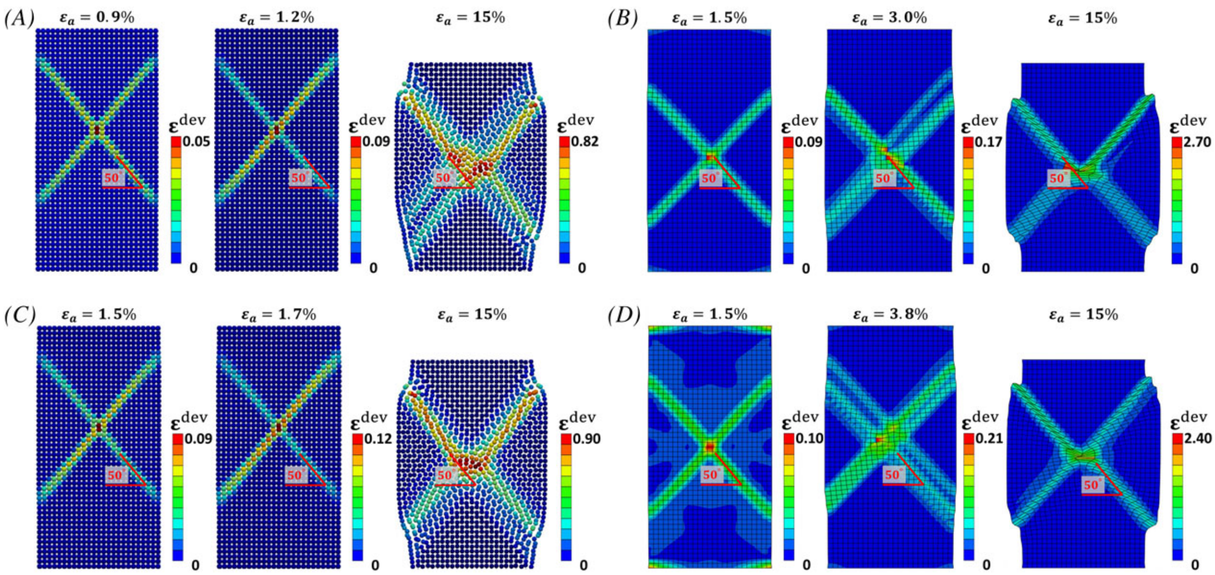

FIGURE 9 Total deviatoric strain plot for biaxial tests with horizontally fixed cap. A, SPH with 50 kPa confinement. B, FEM (with Nlgeom) with 50-kPa confinement. C, SPH with 100-kPa confinement. D, FEM (with Nlgeom) with 100-kPa confinement [Colour figure can be viewed at wileyonlinelibrary.com]

other. The developments of these two shear bands alternate and eventually leads to a strain localisation area with clear X-shaped shear bands at 15% axial strain. Similar process is also observed in the SPH test with 100 kPa confining stress (Figure 9C), which predicts more symmetric X-shaped shear bands. On the other hand, in the FEM tests (Figure 9B,D), an initially smeared deviatoric strain field is also observed. At the bifurcation stress state, a symmetric localisation pattern is also observed. Since the non-linear geometry deformation is considered in FEM (with Nlgeom), the elements within the shear localisation zone rapidly distort as the strain field becomes more localised. When 15% axial deformation is reached, a cross-shaped shear band forms and generally maintains its symmetry. The shear band inclination angle (θ) in both SPH and FEM is measured close to 50° with respect to the horizontal direction, which is close to the empirical relation $\theta = \frac{\pi}{4} + \frac{\phi}{4} + \frac{\psi}{4}$ proposed by Arthur et al.[77] This result suggests that the current SPH framework could produce similar results to that of FEM at small deformation range, while outperforming the conventional FEM method when analysing large deformation soil plasticity problems.

The evolution of deviatoric strain in both SPH and FEM are compared for the free-cap boundary, and illustrated in Figure 10. When the lower confinement is applied (50 kPa), an initial elastic soil response leads to a uniform strain field across the sample. As the stress state is close to the peak point at 1.0% axial deformation (Figure 10A), the deviatoric strain field localises, manifesting as cross-shaped shear bands centred in the sample. At 1.2% axial deformation, the free-cap boundary triggers the strain field to localise into one branch of the cross-shaped shear bands. This leads to a continuous localisation of the deviatoric strain in a single inclined shear band while unloading takes place in the other previously localised areas. When the sample reaches 15% axial deformation (Figure 10A), a clear inclined shear band is observed with significant shear deformation accumulated within this area. A similar development is also observed in the SPH sample with 100 kPa confining stress as demonstrated in Figure 10C. For the FEM comparison test (Figure 10B), a uniform strain field is also observed during the elastic stage of deformation. When the stress state approaches its peak at 1.1% axial deformation, the free-cap boundary instantly triggers the deviatoric strain field to concentrate along the diagonal direction of the meshes. The deviatoric strain continues to localise in this area to form a single inclined shear band while the rest of the sample undergoes elastic behaviour. This localisation band gains its width as deformation grows and maintain its location as the sample achieves 15% axial deformation. The FEM predicted a similar shear band angle to that of SPH and these results are consistent with the empirical relation proposed by Arthur et al.[77] as illustrated in Figure 10. From the above comparisons, the SPH method demonstrates its capability in modelling problems involving extreme soil deformation thanks to its mesh-free nature.

A comparison of axial stress-strain history among SPH, FEM (ABAQUS), and GIMPM[49] methods with fixed and free-cap boundaries is depicted in Figure 11. For fixed cap condition, all models predict an elastic soil response when the sample axial deformation is less than 1%, yet SPH and GIMPM yield an almost identical elastic modulus while FEM

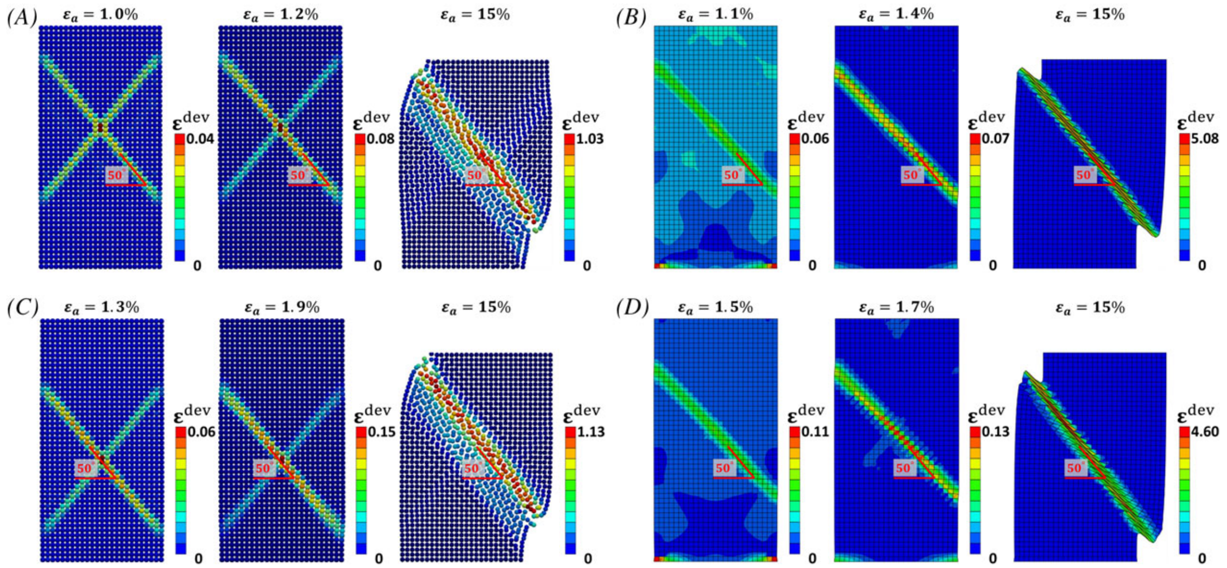

FIGURE 10 Total deviatoric strain plot for biaxial tests with horizontally free cap. A, SPH with 50-kPa confinement. B, FEM (with Nlgeom) with 50-kPa confinement. C, SPH with 100-kPa confinement. D, FEM (with Nlgeom) with 100-kPa confinement [Colour figure can be viewed at wileyonlinelibrary.com]

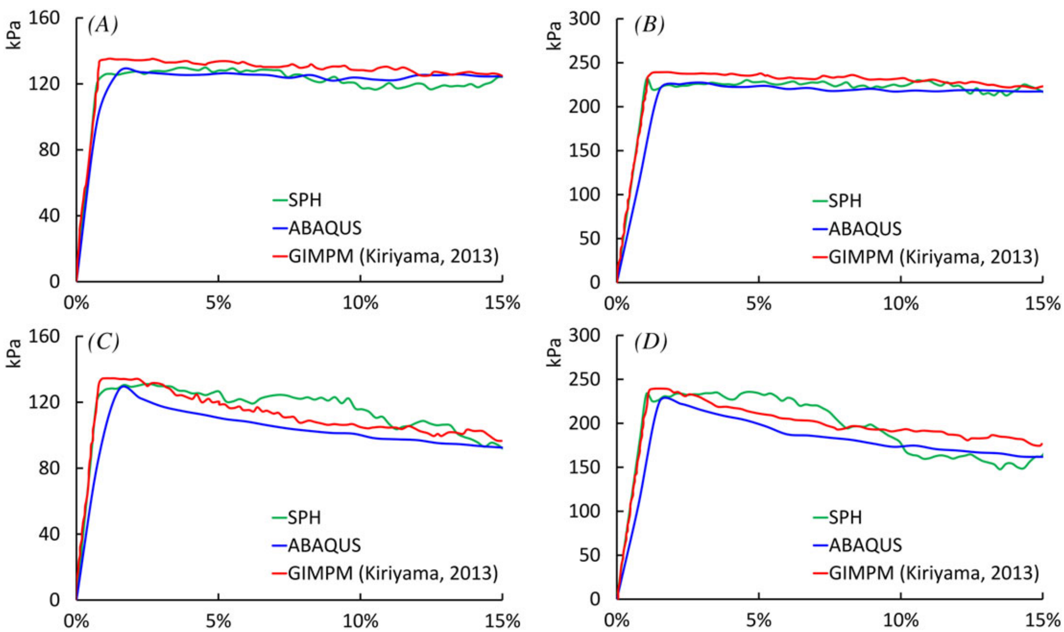

FIGURE 11 Axial stress versus strain of biaxial test from different numerical methods for A, 50-kPa confining pressure with horizontally fixed top, B, 100-kPa confining pressure with horizontally fixed top, C, 50-kPa confining pressure with horizontally free top, D, 100-kPa confining pressure with horizontally free top [Colour figure can be viewed at wileyonlinelibrary.com]

shows small variation. When the stress states in soil approach the peak point, nonlinearity becomes dominant as the increment of stress along with strain reduces and quickly demonstrates a perfectly plastic behaviour in all simulations. A peak stress value of around 125 kPa is predicted by all three methods at this point. These results suggest that SPH could yield similar results to that of FEM and GIMPM and is capable of solving boundary value problems that are consistent with the underlying elastic-perfectly plastic constitutive model. The good agreement among three numerical methods is also observed in the numerical samples with free-cap boundaries. In particular, both SPH and GIMPM again predict an identical elastic modulus, while FEM shows small variation. The SPH results manifest the perfectly plastic behaviour up to 3% axial deformation for 50-kPa confinement and 7% axial deformation for 100-kPa confinement, and then show steady and continuous global softening due to structural failure triggered by the free-cap boundary. A similar pattern is observed in GIMPM results with the perfect plasticity continues up to 2.5% (50-kPa confinement) and 2% (100-kPa confinement) axial strain and then following by softening responses, while FEM depicts an instantaneous softening behaviour after the peak stress point. With free-cap boundary, all models predict their residual stresses around 75% of the peak value as axial strain continues to 15%.

An analysis of varying spatial discretisations in both SPH and FEM is further carried out to verify the reliability of the proposed SPH approach. In particular, the fixed-cap biaxial test under 50-kPa confining stress is repeated with three initial interparticle distances (or mesh sizes in FEM) of dx = 2 mm, dx = 3 mm, and dx = 4 mm. Figure 12 shows a comparison between SPH and FEM for the three lattice sizes. At 2% axial strain, both SPH and FEM predict a rather similar shear banding patterns as shown in Figure 12A,B. The strain localisation configurations are comparable between SPH and FEM for the resolutions of dx = 2 mm and dx = 4 mm, which are evidenced by the symmetric X-shaped shear band configuration observed in both samples. Nevetherless, the sample with dx = 3 mm shows a nonsymmetric X-shaped shear configuration in the FEM solution, while SPH shows competing mechanisms between the symmetric and nonsymmetric shear bands. The nonsymmetric results can be attributed to the influence of spatial discretisations (ie, mesh discretisations in FEM) as no weak zones were defined in the numerical samples to trigger localised deformation. However, owing to the SPH nature of particle approximations and probably the perfectly plastic MC model used in this study, its solutions seem to mitigate the influence of spatial discretisations. This explains why the SPH results of dx = 3 mm show competing mechanism between symmetric and nonsymmetric ones. Nevertheless, both methods predict the same shear band orientation, which is about 50^{o} to the horizontal direction. As the axial deformation reaches 15%, the localisation of deformation continues to develop, leading to thickening of shear bands. The SPH results show comparable symmetric X-shape shear bands among three numerical samples (Figure 12C), thanks to the particle approximation in SPH. In contrast, the final shear band configurations predicted by FEM are rather different among three numerical samples, both in terms of their symmetry and location (Figure 12D), and this can be attributed to the influence of both mesh discretisations and distortions. The comparison of axial stress-strain relations between SPH and FEM for the three tests are shown in Figure 13. Both SPH and FEM predict nearly identical stress-strain curves for different spatial discretisations (or meshes in FEM) and the results are comparable between two methods. This similarity can be attributed to the perfectly plastic model adopted in the current simulations, which enforce the stress to stay on its yield loci during the plastic deformation.

The above results demonstrate that the SPH method can produce comparable results to those of the classical mesh-based FEM for small deformation and GIMPM for large deformation. The proposed confining boundary condition for SPH has been shown to be stable and is able to maintain the prescribed confining stress at very large deformation.

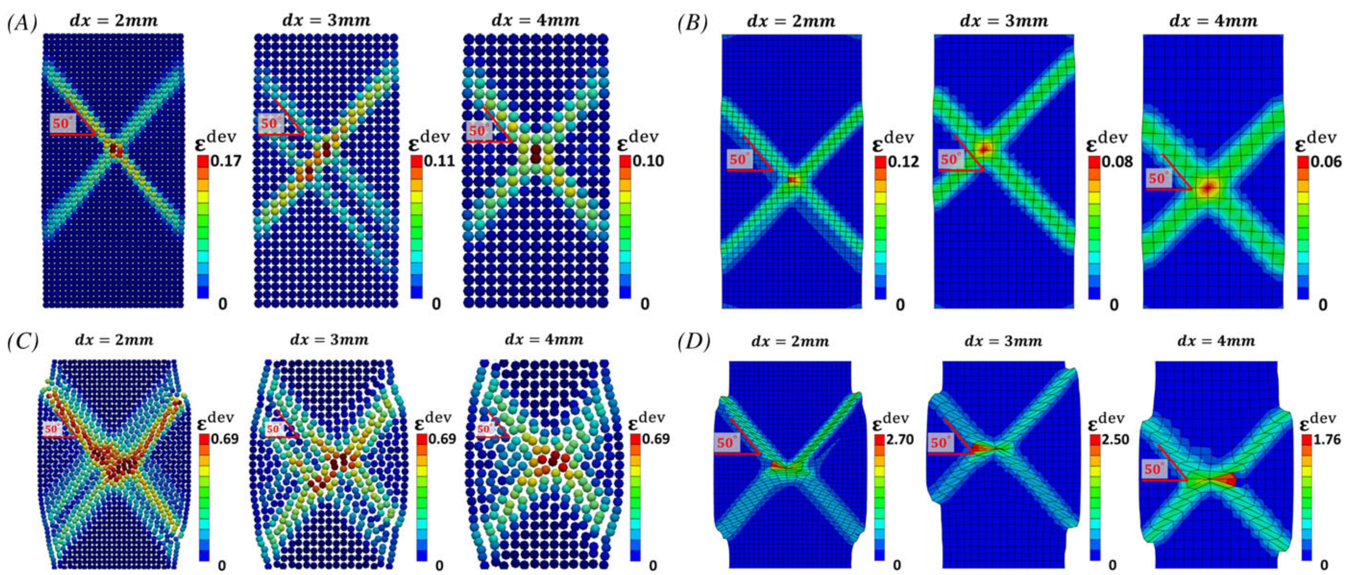

FIGURE 12 Total deviatoric strain plot for different domain resolutions (2, 3, 4 mm). A, 2% axial strain in SPH. B, 2% axial strain in FEM. C, 15% axial strain in SPH. D, 15% axial strain in FEM [Colour figure can be viewed at wileyonlinelibrary.com]

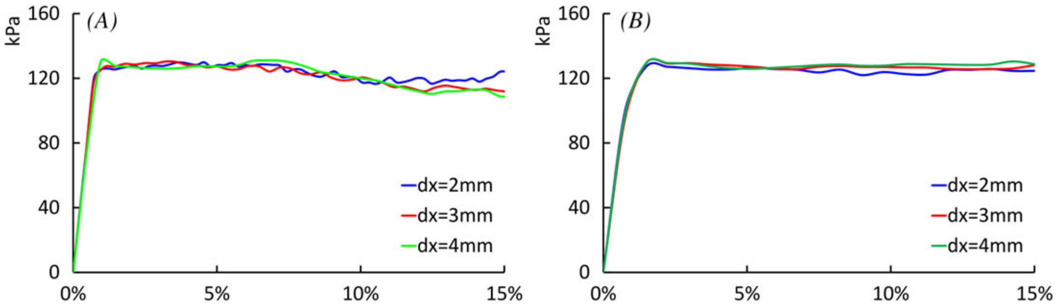

FIGURE 13 Axial stress-strain relations among varying resolutions. A, SPH predictions. B, FEM predictions [Colour figure can be viewed at wileyonlinelibrary.com]

The extension of this approach to three-dimensional applications and its predictive capability will be examined in the next section.

5.3 | SPH simulation of the triaxial test

The proposed SPH framework has been compared with FEM and GIMPM solutions for plane strain soil plasticity problems and very good agreements among three numerical approaches have been achieved. In this section, the SPH framework is further extended to the three-dimensional application by conducting simulations of triaxial tests and numerical results are compared with triaxial experimental data reported by Kiriyama.[49] The performance of the proposed confining boundary condition is also examined under three-dimensional condition. The geometry and boundary conditions of this test are shown in Figure 14 together with the SPH discretisation of the numerical sample. The cylindrical specimen has a dimension of 25 mm in radius and 100 mm in height. Fan-shaped particle lattice is utilised to discretise the numerical sample with the aim to create a symmetric particle configuration in both cross-sectional and axial directions, thus minimising heterogeneous effects associated with the orthogonal particle discretisation. A total number of 28 140 SPH particles are used to create the numerical sample with an initial interparticle distance of 2 mm. The loading platens are represented by five extra layers of particles created at top and bottom of the soil domain. Fixed-cap and free-cap boundary conditions are imposed at the top loading platen, while the bottom one remains fully fixed during the computation. The triaxial loading test is started by applying a constant downward speed of 10 mm/s to the top loading platen. Soil properties are taken from Kiriyama[49] and other SPH parameters are set similar to those utilised in the biaxial tests, which are all listed in Table 1.

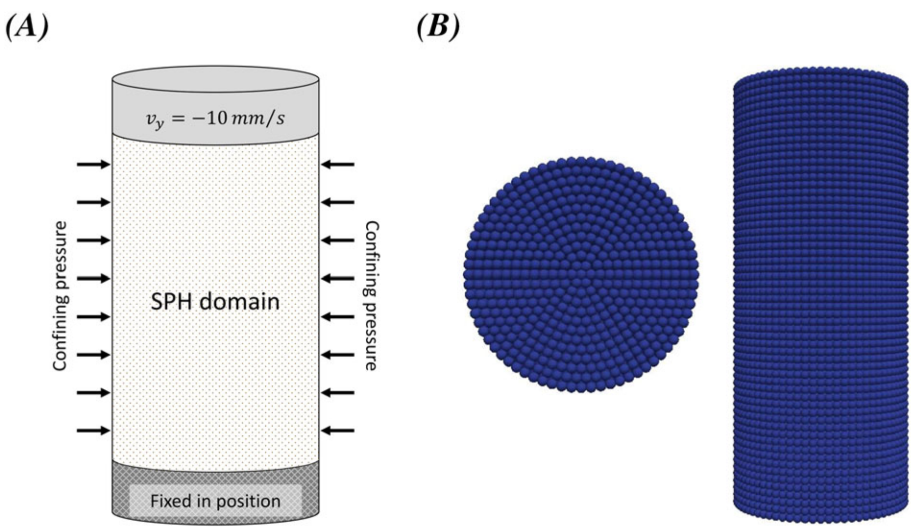

FIGURE 14 Triaxial test numerical set-up in SPH. A, Geometry and boundary condition. B, SPH discretisation of the sample [Colour figure can be viewed at wileyonlinelibrary.com]

The evolutions of localised strain field with fixed-cap triaxial tests under 50- and 100-kPa confining stresses are illustrated in Figure 15. The shear bands are observed in both perspective and cross-sectional views. In the test with 50-kPa confining stress (Figure 15A,C), the strain field is distributed across the sample at an early stage of elastic deformation. Immediately after the stress passes its peak point at 5% axial strain, the deviatoric strain field localises into symmetric X-shaped shear bands. As the axial deformation continues, the localisation of deformation becomes more apparent. Upon 15% axial strain, cross-shaped shear bands are observed, while multiple nonsymmetric bands are formed at the centre of the soil cylinder (Figure 15C). The shear band evolution at 100-kPa confining stress is analogous to that at 50-kPa confining stress. However, once symmetric shear bands are formed inside the soil sample, their configuration and location are maintained during the plastic deformation. As 15% axial strain is achieved, shear bands demonstrate an increase in width while its symmetry, to some extent, is maintained (Figure 15B,D). For the test with free-cap boundary condition under 50-kPa confinement, the shear band development is shown in Figure 16A,C for 5% and 15% axial deformations, respectively. At 5% axial deformation, the free-cap boundary triggers multiple crossed and inclined shear bands. As deformation continues, the free-cap boundary facilitates a fast accumulation of shear deformation, and the shear bands start to localise into one branch with the outside zone undergoes unloading. Upon 15% axial deformation, a single inclined localisation band remains in the sample. For the test with 100-kPa confinement, the sample undergoes a similar process, yet the shear deformation is more concentrated, manifesting a single shear band with smaller thickness compared with that of the 50-kPa confinement case. As for the shear band angle, tests with both fixed and free top boundaries predict the same inclining shear band angle of 50° as shown in Figures 15 and 16. This result is consistent with the experimental observation reported by Kiriyama.[49]

Figure 17 shows a comparison of axial stress-strain relationships obtained by three numerical methods (SPH, FEM, and GIMPM) and experiments. For the simulations with fixed-cap boundary under both 50- and 100-kPa confinements, the stress-strain curves exhibit an overall linear elastic-perfectly plastic behaviour, and these results agree well with the experimental data. All simulations yield constant vertical stresses at around 120 and 220 kPa for the low and higher

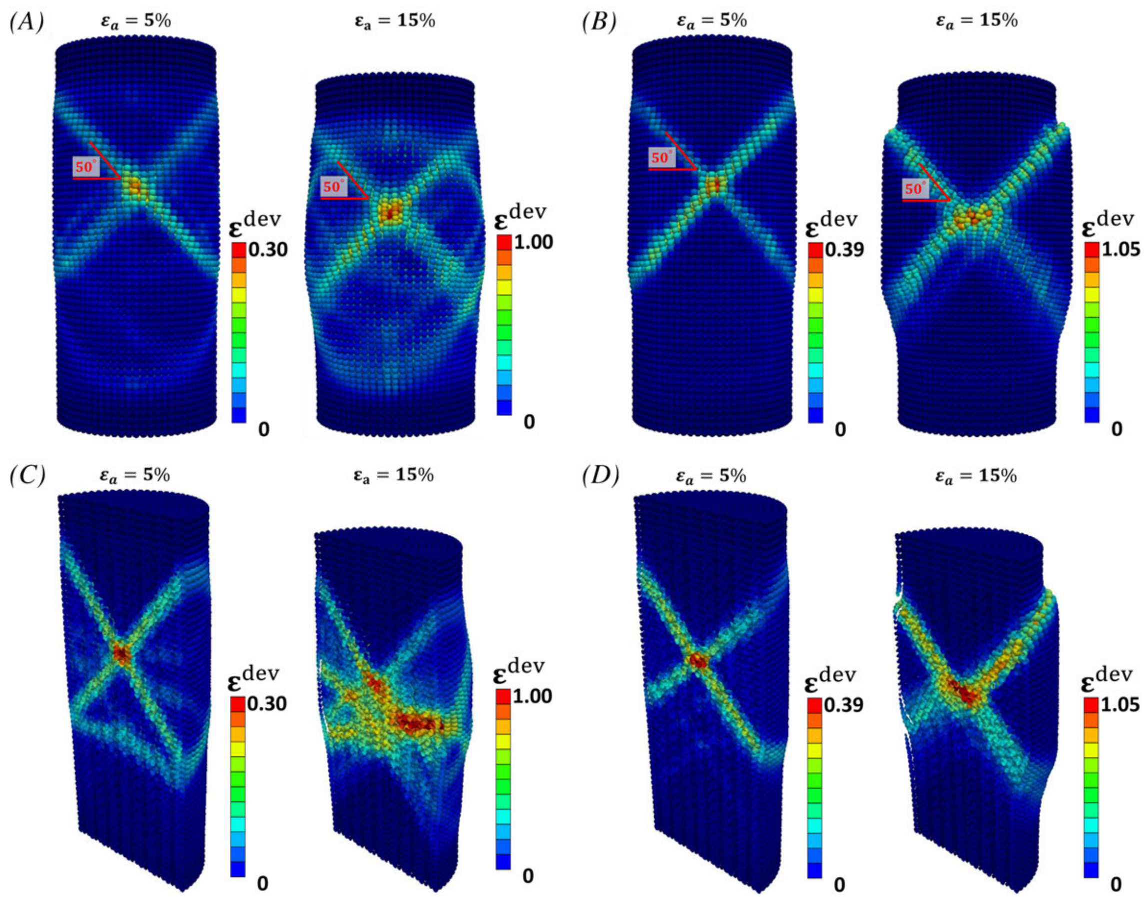

FIGURE 15 Total deviatoric strain in triaxial tests with fixed cap condition. A, 50-kPa confining stress. B, 100-kPa confining stress. C, Cross-sectional plot of 50-kPa confining stress. D, Cross-sectional plot of 100-kPa confining stress [Colour figure can be viewed at wileyonlinelibrary.com]

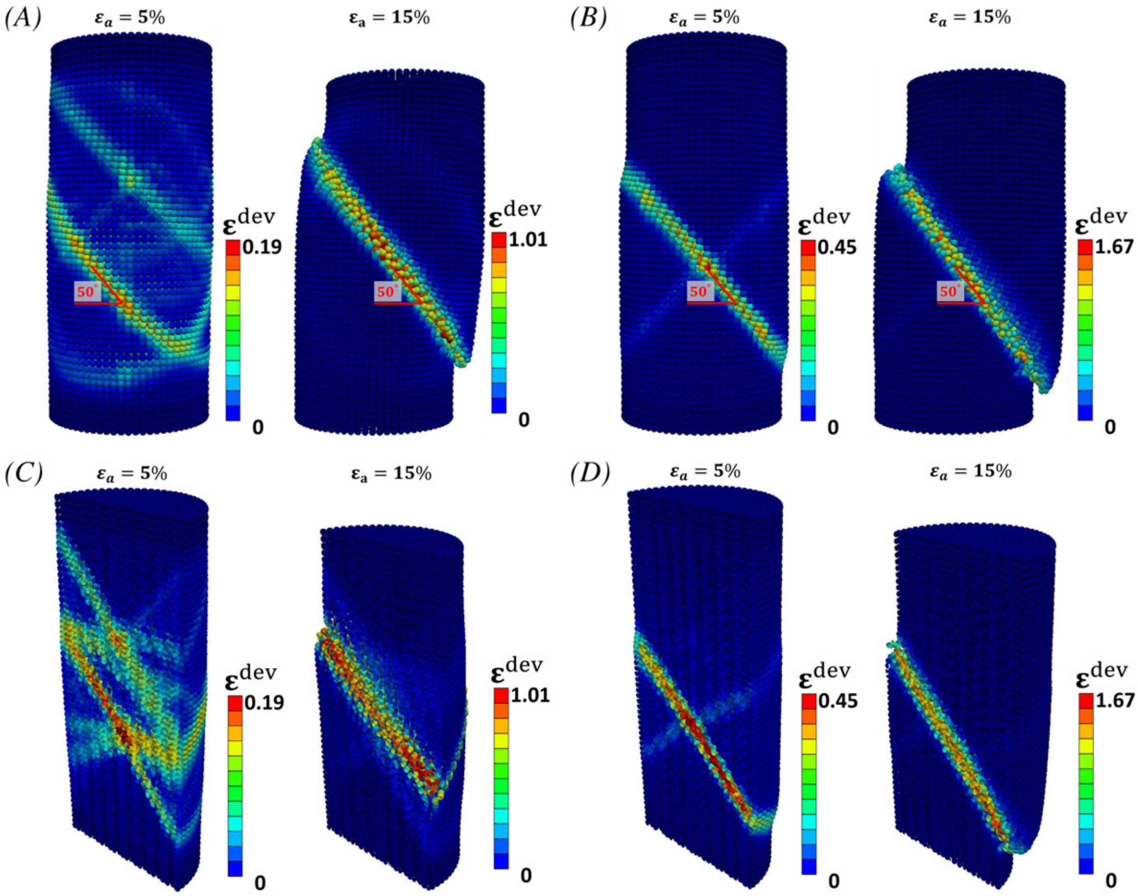

FIGURE 16 Total deviatoric strain in triaxial tests with free cap condition. A, 50-kPa confining stress. B, 100-kPa confining stress. C, Cross-sectional plot with 50-kPa confining stress. D, Cross-sectional plot with 100-kPa confining stress [Colour figure can be viewed at wileyonlinelibrary.com]

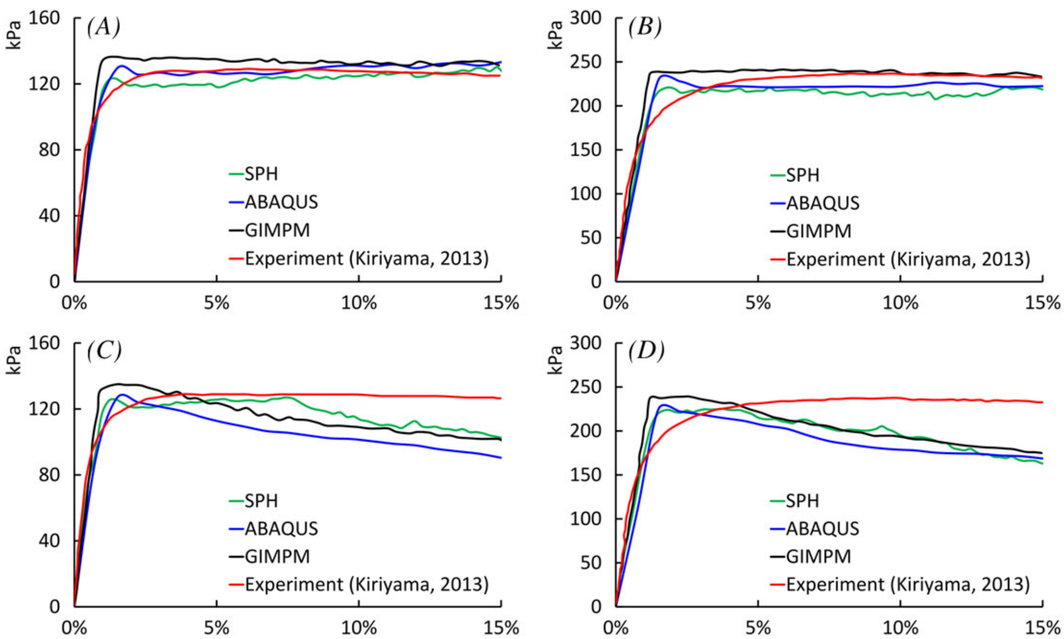

FIGURE 17 Axial stress versus axial strain of triaxial test from different numerical methods for A, 50-kPa confining pressure with the horizontally fixed top. B, 100-kPa confining pressure with the horizontally fixed top. C, 50-kPa confining pressure with the horizontally free top. D, 100-kPa confining pressure with the horizontally free top [Colour figure can be viewed at wileyonlinelibrary.com]

confinement cases, respectively, and these results are again in good agreement with the experimental data reported by Kiriyama.[49] For the cases with free-cap boundary, as the stress state passes the peak point, SPH and GIMPM demonstrate a short period of perfectly plastic behaviour. For example, at 50-kPa confinement, the perfect plasticity continues up to 7.9% and 3.6% axial deformation in SPH and GIMPM results, respectively. This is then followed by a persistent softening curve predicted by both methods. FEM yields persistent softening immediately after the peak stress. However, the experimental data demonstrate perfectly plastic curve until 15% axial strain. The possible explanation for this phenomenon is due to the fact that the strictly free-cap boundary condition is hardly achieved in the laboratory. Accordingly, the perfectly plastic behaviour in soil maintains up to 15% axial deformation. In contrast, the free-cap condition is easily achieved in the numerical simulations, which facilitates shear localisation and leads to the structural failure thus softening behaviour in the sample.

Finally, to further demonstrate the effectiveness of the proposed confining boundary condition in maintaining the correct loading condition and its effectiveness of treating the corner singularities in the Mohr-Coulomb model, the local loading path of a particle located at the centre of the sample and the global loading path measured from the sample boundaries are plotted in Figure 18, on both meridian and deviatoric planes. Both loading paths start from point O, corresponding to the initial condition, where the sample is under hydrostatic pressure of 50 kPa. As the axial load on the top surface starts increasing, the local loading path at the measuring point develops linearly until reaching point A and the sample is relatively homogeneous evidenced in the coincidence of local and global loading paths. At this stage, the deviatoric plastic strain is zero everywhere inside the sample. The slope of the local loading path OA measured in Figure 18A is $1/\sqrt{3}$, which coincides with the global triaxial loading path imposed on the sample shown as the black dashed line. As the local loading path goes past point A, some particles inside the sample enter their plastic state (evidenced by the development of deviatoric plastic strain at point B in Figure 18A). This leads to a slightly nonlinear section of the local loading path between point A and point C on the meridian plane. However, the global loading path measured at the boundaries continuously shows a linear behaviour, indicating that the triaxial loading condition over the entire sample is well maintained through the proposed boundary condition. This is also evidenced by the straight loading paths (both local and global) that evolve exactly towards the triaxial compressive meridian in the deviatoric plane (Figure 18B). Beyond the initial yield at point C, the local stress just stays on the yield envelope and moves from C to D in Figure 18A while still lying at the corner of the Mohr-Coulomb yield surface in Figure 18B. This confirms that the consistency condition is well satisfied at the measuring point, and the triaxial loading path imposed on the sample is well maintained by the proposed confining boundary condition, even during the post yielding process. It is also noted that the loading path CD in Figure 18B corresponds to the triaxial compressive meridian on the Mohr-Coulomb yield surface, where the corner singularity issue arises. However, no numerical instability is observed in our simulations, suggesting that the proposed algorithm is effective in treating the corner singularity of the Mohr-Coulomb model, which maintains a valid flow rule during material plastic deformation.

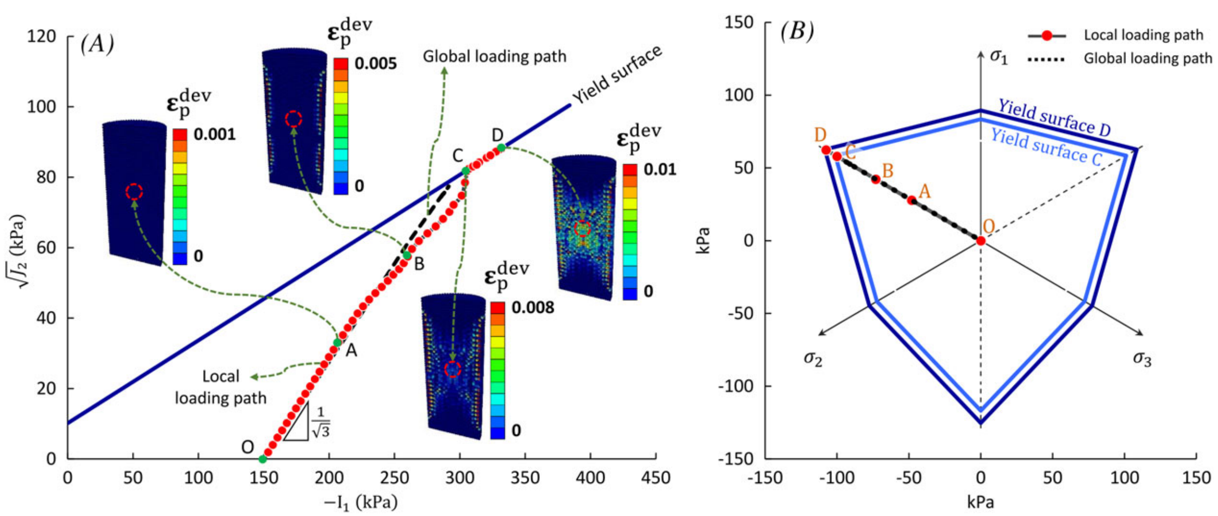

FIGURE 18 Global loading path and a local one of a particle located at the centre of the sample under 50-kPa confinement. A, Meridian plane. B, Deviatoric plane [Colour figure can be viewed at wileyonlinelibrary.com]

The above numerical results verify that the proposed confining SPH boundary works well in both two- and three-dimensional conditions. The method is highly stable and is able to maintain the confining stress on highly non-linear curvature confining boundaries caused by excessively large soil deformation. Although a constant confinement scenario is considered for above tests, the proposed boundary method works equivalently for evolving confinement and can be applied, in general, for any SPH problems involving confinement. The Mohr-Coulomb perfectly plastic model when incorporated in SPH show comparable results to those of FEM for small deformation, GIMPM for large deformation and experiment. This demonstrates the capability of SPH in handling general soil plasticity problems while outperforming the traditional FEM-based methods in applications involved large deformation and failure. There still remains questions on the dependency of the solutions on SPH discretisation. The current results in the biaxial and tri-axial tests show less dependence of numerical results on the domain discretisation, particularly in terms of load-displacement curves. This is not a true evidence to demonstrate mesh-independency of SPH given a perfectly plastic model was used. The fixed yield surface may regularise the energy dissipation during plastic deformation and lead to plausible mesh-independency of the method. To have further insights into this problem, a softening constitutive model is required. This is beyond the scope of the current paper and a subject of our ongoing investigations.

6 | CONCLUSION

In this article, a generic method is proposed to effectively enforce flexible confining boundary conditions in SPH. The method makes use of its kernel truncation feature along free edges of SPH domain to automatically locate and trace curvature change of the boundary surfaces. The confining stress and its normal vectors are subsequently the natural outcomes of SPH approximations of the kernel gradient of confining stress for those SPH particles located on or close to the confining boundaries. This method has demonstrated its capability to effectively enforce a constant confining stress on to flexible boundaries, although in principle it works equivalently for evolving confinement. In parallel with this, a robust numerical procedure for implementation of the Mohr-Coulomb elasto-plastic model in SPH, which avoids numerical difficulties associated with the well-known singularities that occur at the corners of the Mohr-Coulomb surface, is presented for the first time. The proposed numerical framework was first verified with theoretical solutions under simple shear condition. It was then used to simulate biaxial and triaxial tests with varying model parameters and numerical results were compared with those from the mesh-based method (FEM), advanced hybrid particle-mesh method (GIMPM), and experimental data available in the literature. Very good agreements between SPH and other counterparts were achieved, suggesting that the proposed confining boundary condition fulfils the need of SPH for general geotechnical applications that involved flexible and moving confining boundaries. The good agreement between SPH and experiment also suggests that, once a suitable constitutive model is selected and properly implemented in SPH, the method can be used to predict complex problems involved large deformation and failure. For this purpose, more advanced constitutive models capable of capturing realistic soil behaviour should be implemented and tested with the SPH method. These are beyond the scope of this paper and will be left for future works.

ACKNOWLEDGEMENTS

This work was supported by the Australian Research Council through Discovery Projects DP160100775 (Bui), FT140100408 (Nguyen), and DP170103793 (Nguyen and Bui). Funding support from CISRO to the first author is also gratefully acknowledged.

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How to cite this article: Zhao S, Bui HH, Lemiale V, Nguyen GD, Darve F. A generic approach to modelling flexible confined boundary conditions in SPH and its application. *Int J Numer Anal Methods Geomech*. 2019;43:1005–1031. https://doi.org/10.1002/nag.2918